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Timestep-Conditioned Adapter Scaling for Multi-view Diffusion Texture Generation: Control Allocation and Baked Appearance

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ABSTRACT

Multi-view diffusion pipelines for Geometry-conditioned multi-view diffusion generates images that are subsequently baked onto 3D asset texturing commonly inject geometric conditions, such as normal and depth maps, through adapter modules, and the adapter scale at inference time is usually chosen empirically, although the generated views are later baked into the final mesh texture. We show that a single uniform scale is a blunt control: on the studied adapter, aggressive uniform scaling improves structural metrics such as foreground structural similarity (FG-SSIM) but suppresses the texture synthesis capability of the base model, causing flattened surfaces, reduced color variation, and loss of high-frequency details. We further identify an adapter-specific temporal conflict: geometric residuals are most useful for mid-stage structure refinement, whereas strong residual injection in the early or late stages disturbs global formation or suppresses texture synthesis. We introduce a measurable stage-utility analysis with structural guardrails and derive meshes, yet the benefit of time-varying geometry control depends on both the implemented intervention and the resulting textured asset. We revisit Timestep-Conditioned Adapter Scaling (TCAS), a training-free TCAS) on an existing GeoTex residual interface, distinguishing requested control strength, native-cap-adjusted applied scales, and state-dependent residual corrections.

On 300 held-out objects, the historical low–high–low schedule that concentrates strong geometric correction in the middle denoising stage and uses conservative scales otherwise. The schedule C3 $= (1.25, 2.50, 1.25)$ is selected using only a 24-object probe set from 17 scaling variants and then evaluated unchanged, without further search, on the strictly disjoint 276-object holdout. On the full 300-object pooled evaluation with the main adapter, TCAS achieves FG-SSIM comparable to aggressive uniform scaling while improving the peak signal-to-noise ratio (PSNR) by 0.96 dB; it also retains substantially more texture and receives higher preference in a blinded human study. These results suggest treating adapter strength as a stage-dependent inference control variable within the tested MVPainter-style architecture family rather than as a fixed hyperparameterprofile improves foreground PSNR over Fixed Low by 0.522 dB, but this comparison also increases the applied control budget. Stronger fixed and linear alternatives achieve better mean scores across the four evaluated image endpoints. When per-group applied-scale sums are matched, the specified middle-window contrast is -0.0339 dB (95% interval $[-0.1172, +0.0514]$), providing no detected advantage for the proposed middle-window allocation. A separate retrospective pairwise analysis distinguishes the middle profile from Early-high and Late-high in both FG-PSNR and FG-LPIPS within this cohort, without establishing a preferred placement.

Native-UV baking reveals a small Late-high unseen-view LPIPS improvement on ten compatible GeoTex objects (-0.00265), while substantial reference-appearance error remains. A separate twenty-object bake study does not detect the same benefit, and native conditioning-interface comparisons do not establish a transferable middle-window rule.

These results clarify how geometry-control allocation should be evaluated through its executed interface, competitive alternatives, and final textured-asset appearance. Rather than identifying a universally preferred schedule, the study provides task-specific evidence about the effects and limits of inference-time geometry control in multi-view texturing.

1. Introduction

Recent advances in single-image 3D generation have substantially improved the recovery of geometry, multi-view consistency, and sparse-view reconstruction [4, 5, 6, 7, 8, 9, 10, 11]. As geometric generation models become increasingly mature, generating high-quality textures on an existing 3D shape has become a key factor affecting the realism and usability of 3D content. For an object whose geometry has already been determined, the final visual quality depends not only on whether the shape is complete, but also on whether the texture is faithful to the reference image, aligned with the 3D surface, and rich in local details.

Existing multi-view texture generation methods usually generate RGB images from multiple target views using 2D image diffusion models, and then map the generated images back to the A textured 3D surface through projection or texture baking [13, 14, 15, 16]. This route benefits from the generative capability of image diffusion models, but it still faces challenges in reference appearance alignment, geometry-texture consistency, and local texture detail preservation. Geometry-controlled asset must preserve recognizable appearance from cameras that were not used to generate or project its texture. Geometry-conditioned multi-view diffusion methods represented by MVPainter introduce geometric conditions such as normals and depth maps to improve the alignment between generated textures and the underlying 3D surface, demonstrating the importance of explicit geometric control for 3D texture generation [22]. diffusion produces RGB views that are transferred onto a mesh, but the images are intermediate outputs: the graphics result is the textured asset after projection, coverage, and resampling.

However, introducing geometric conditions is not only a question of whether a control signal should be used, but also a question of how strongly it should be applied. In adapter-augmented diffusion pipelines, geometric conditions are usually injected into the model through an adapter or control branch, and their influence is regulated by a scaling factor. In practice, this factor is often set empirically: a small scale may lead to insufficient geometric constraints, whereas a larger scale is usually expected to improve structural consistencyScaling an existing geometry adapter provides an inference-time control that can also vary during denoising. Yet a low–high–low profile compared with a weak constant changes both control placement and total strength. Native wrapper caps may further clip requested values differently across injection groups, while the resulting correction depends on the evolving latent state. Neither the requested curve nor a matched scalar sum alone identifies a temporal-placement effect.

In multi-view texture generation, however, stronger adapter scaling is not necessarily better. Texture generation requires not only contours and structures to be consistent with geometry, but also surface color variations, material details , and high-frequency patterns to be preserved. Overly strong geometric control may force the model to follow normal, depth, or edge conditions too aggressively, thereby suppressing the intrinsic texture synthesis capability of the base diffusion model. In this case, some structural metrics may still improve, while the visual result suffers from texture flattening, reduced color variation, and detail lossImage-space agreement is also distinct from final appearance. Reference-color changes and missing texture details can coexist with improved view metrics; projection and resampling add further differences. Texture-variation statistics can rise because of artifacts as well as detail. Evaluating generated views alone therefore cannot establish how a UV-textured asset appears from unseen cameras.

Therefore, adapter scaling in geometry-controlled multi-view diffusion deserves further analysis. Compared with prior work that mainly studies how to build texture generation pipelines, how to introduce geometric conditions, and how to improve overall generation quality, adapter strength is a fine-grained inference-time control problem. It directly affects the balance between shape consistency and texture detail, and is an easily overlooked factor in multi-view texture generation quality.

This problem is related to, but not solved by, existing guidance schedulingstrategies. CFG scheduling changes the extrapolation between conditional and unconditional predictions, while a geometry adapter injects residuals into intermediate representations. Generic ControlNet or adapter scale scheduling similarly treats condition strength mainly as a condition-adherence knob. In texture generation, however, the key question is not only how strongly to follow the geometry, but also when geometric residuals should dominate and when they should recede so that the base model can synthesize color, material, and high-frequency texture. Simple monotonic schedules such as linear or cosine decay do not explicitly address this non-monotonic requirement. Even the recent finding that classifier-free guidance is most useful only in a limited noise interval [36] does not answer this question: it identifies where conditioning helps for CFG, but not which stages of a geometry-conditioned texture pipeline require strong geometric residuals, where the late-stage risk is texture suppression rather than diversity loss. We revisit Timestep-Conditioned Adapter Scaling (TCAS) on an existing GeoTex residual interface, using the recorded C3 operating point $(1.25, 2.50, 1.25)$ over early, middle, and late denoising windows. Layer and timestep conditioning precedents, including ControlNet controls and Scheduled Style Injection, are established [1, 2, 3]; we do not claim to introduce temporal scheduling. We ask whether C3 remains competitive against strong fixed and linear controls, what can be attributed to relocating matched applied-scale budgets, and whether the differences persist in native-UV bakes or other conditioning interfaces.

The goal of this paper is not to propose a new full texture generation system, but to analyze and improve inference-time adapter scaling within a fixed geometry-controlled multi-view diffusion pipelineThe results revise the interpretation of C3. It improves foreground PSNR over Fixed Low on 300 held-out objects, but Fixed High and Generic Linear have better means across the four image endpoints. The specified matched-budget primary does not detect a middle-window advantage; a separate retrospective pairwise analysis distinguishes the middle placement from both early and late placements within this cohort, without establishing a preferred schedule. Native-UV baking finds a small Late-high improvement at unseen cameras alongside substantially larger reference-appearance error; a separate bake protocol does not detect the same benefit. Native MV-Adapter comparisons likewise do not establish a transferable middle-window rule.

The main contributions of this paper are as follows:

(1) We systematically diagnose the shape-texture trade-off caused by adapter scaling in multi-view diffusion texture generation, showing that higher scales can improve structural metrics while simultaneously causing texture degradation whose specific form (flattening, artifact amplification, or no effect) depends on the adapter 's residual magnitude and scale semantics.

(2) We identify an adapter-specific temporal conflict between geometric residual injection and texture formation, and propose Timestep-Conditioned Adapter Scaling (TCAS), which concentrates strong geometric correction in the middle denoising stage while using conservative scales in the early and late stages.

(3) Through uniform scale sweeps, texture audits, a 24-object probe analysis of 17 uniform, layer-wise, and temporal scaling variants, and a 300-object evaluation pool without any further schedulesearch, we show why fixed high scales and simple stage transfers of guidance scheduling are insufficient for the shape-texture trade-off.

(4) We further provide a large-scale texture preservation analysis on the 300-

object evaluation pool, showing that TCAS substantially reduces texture flattening compared with full high-scale scaling: contributions are threefold. **First, implementation-aware characterization of geometry-control allocation:** we distinguish requested, cap-adjusted applied, and state-dependent residual control on an existing GeoTex adapter interface, making clear which intervention is executed and which differences cannot be attributed to temporal placement alone. **Second, controlled evaluation of strength and temporal placement:** we compare historical C3 outperforms $s = 2.50$ on 87.7%, 95.0%, and 83.0% of objects in Laplacian variance, RGB standard deviation, and gradient magnitude, respectively, while the 95% confidence interval for the FG-SSIM difference crosses zero, indicating no significant difference in structural similaritywith competitive fixed and linear controls and with schedules matched in per-group applied-scale sums, revising the interpretation of C3's gain over a weaker fixed setting and showing within-cohort allocation sensitivity without identifying a preferred schedule. **Third, asset-level assessment and transfer boundaries:** we evaluate native-UV textured meshes against matched reference self-bakes from unseen cameras and examine related scheduling questions within separate native conditioning interfaces. Together, these experiments connect generated-view measurements to final-asset appearance while identifying protocol- and endpoint-dependent limits. This is a task-specific empirical contribution, not a general schedule selector or an independently derived optimum.

2. Related Work

2.1. 3D Texture Generation and Multi-view Diffusion

Single-image 3D generation is often decomposed into geometry generation and texture generation. In recent years, single-image-to-3DSingle-image-to-3D, multi-view diffusion, and sparse-view reconstruction methods have made notable progress in structural stability , shape completeness, and cross-category generalization [4, 5, 6, 7, 8, 9, 10, 11]. Nevertheless, high-quality texture generation remains a key bottleneck for the improved structural stability and view synthesis [4, 5, 6, 7, 8, 9, 10, 11]. Texture synthesis remains a distinct challenge for asset realism and usability of 3D content, as reflected by recent surveys of neural 3D mesh texturing [12]. Existing texture generation methods usually rely on [12]. Many systems generate RGB views with 2D diffusion models to synthesize RGB images from multiple views , and then diffusion and project or bake these images onto the 3D them onto a surface [13, 14, 15, 16]. While this strategy inherits the strong generative capability of image diffusion models, it still suffers from reference appearance misalignment, geometry-texture inconsistency, and insufficient local texture detail. Beyond these diffusion pipelines, the graphics community continues to advance surface appearance synthesis more broadly, from Recent work also explores exemplar-based texture synthesis [17] to and diffusion-based procedural material generation [18]materials [17, 18].

To address these issues, methods such as Paint3D, MVPaint, Hunyuan3D 2.0, and MVPainter formulate 3D texture generation as a geometry-controlled multi-view diffusion task, generating use geometry-conditioned multi-view images under geometric control and baking them into UV textures generation in 3D texturing pipelines [19, 20, 21, 22]. MV-Adapterfurther shows that adapter-style multi-view control can enhance cross-view consistency without fully retraining the base model [23], and recent texturing systems strengthen consistency during texture synthesis itself, for example through , MVDiffusion, and other synchronized multi-view diffusion [25] or consistent-and-seamless text-guided multi-view generation [26]. By injecting geometric information such as normals and depth maps into the diffusion model as conditional signals, the generated results better follow the 3D surface structure and remain consistent across views. Unlike prior methods that focus mainly on multi-view image generation or general 3D reconstruction, these works emphasize precise alignment between texture and geometry as well as local detail recovery, significantly improving 3D texture generation quality. However, they mainly answer how to introduce geometric conditions and how to construct a texture generation pipeline, while the influence of geometric control strength at inference time on shape and texture quality remains underexploredmethods address consistency through different adapter and denoising designs [23, 24, 25, 26]. These systems establish the graphics context for this work. The present evidence does not compare final baked-texture quality across those systems.

2.2. Adapter-Augmented Diffusion Models and Geometric Control

Adapter modules have become an important mechanism for introducing additional conditions into diffusion models. ControlNet and T2I-Adapter and ControlNet inject spatial conditions such as edges, depth maps, and normals into a pretrained diffusion model through lightweight parallel branches with a controllable scaling factor [27, 28]provide learned spatial-conditioning branches or feature injections, whose contribution can be weighted by an inference-time condition-strength scale [27, 28]. They establish conditioning strength as a practical control, but their injected tensors and scaling sites do not share a numerical dose with GeoTex. The studied GeoTex-Adapter path already exposes an inference-time residual scale and evaluates fixed requests. TCAS reuses that control point as a three-stage request over sampling progress, while recording how wrapper-group caps change the applied values and how adapter activations determine the resulting corrections. It adds no learned branch or schedule selector. IP-Adapter introduces image prompts through decoupled attention [29]. LoRA and its variants implement lightweight model adaptation through ; LoRA and related methods use scaled low-rank residual updates [30, 31, 32]. Although these methods differ in architecture, they generally include some form of scaling factor that controls the influence of the external condition or adapter residual on the base model.

In practice, the adapter scale is often treated as an empirical hyperparameter. A larger scale usually improves condition adherence, for example by better following normals, depth maps, or reference images. In texture generation tasks, however, overly strong conditional constraints may weaken the base diffusion model's ability to synthesize color, material, and high-frequency details. Adapter scaling is therefore not a simple matter of increasing the strength as much as possible; it involves a balance between geometric constraints and texture fidelity. This paper focuses on exactly this inference-time control problem: given a fixed adapter structure and fixed geometric conditions, how should the adapter strength be set or scheduled to preserve both shape consistency and texture detail?

2.3. Diffusion Guidance Scheduling and Timestep Control

The denoising process of diffusion models is highly stage-dependent. Early steps mainly determine global layout and coarse structure, middle steps refine shape and semantic structure, Diffusion denoising is often discussed in terms of coarse-to-fine evolution [33, 34]. Classifier-free guidance combines conditional and late steps are more responsible for color, texture, and high-frequency details [33, 34]. Based on this observation,

many guidance scheduling strategies have been explored in both research and practice. For example, classifier-free guidance scales may be adjusted linearly, with cosine schedules , or dynamically to mitigate oversaturation, detail loss, or reduced diversity caused by overly strong guidance [35, 37]. In addition, the development of latent diffusion, high-resolution diffusion, and Transformer-based diffusion models further shows the close relationship among generation quality, noise schedules, conditional injection, and network architecture *[removed citation:* `rombach2022ldm esser2024scaling blackforestlabs2024flux loshchilov2019decoupled cherti2023reproducible nichol2022glide saharia2022photorealistic peebles2023scalable`*]*. unconditional score predictions [35]; limiting it to an interval studies quality and diversity [36]. Adaptive Guidance instead schedules conditional/unconditional evaluations for inference efficiency [37]. These score-space and computation schedules differ from the geometry-residual intervention tested here.

These methods all exploit the staged nature of denoising to regulate conditional strength. Notably, Kynkäänniemi et al. show that classifier-free guidance is harmful at high noise levels, largely unnecessary at low noise levels, and beneficial mainly in a middle interval, motivating limited-interval guidance [36]; mid-stage concentration of conditioning strength therefore has a close precedent in CFG scheduling . However, different control signals act through different mechanisms. CFG scheduling adjusts the weight between conditional and unconditional predictions, mainly affecting the balance among semantic control, diversity, and visual fidelity. By contrast, the scale of a geometric adapter acts directly on intermediate features or residual branches, controlling how strongly geometric conditions intervene in the generation process. TCAS differs from limited-interval guidance in both mechanism and target: it scales adapter feature residuals rather than score extrapolation, raises the middle scale above a conservative baseline instead of switching guidance on and off, and addresses the shape–texture trade-off of Earlier official ControlNet code already exposes layer-dependent control weights, and Diffusers v0.21.0 documents temporal conditioning start/end gates [1, 2]. SSI is the closest combined layer/time scheduling precedent. It varies StyleID's query-mixing parameter [38] across decoder layers and denoising time, tests schedule direction and shape, and also schedules a separate geometry-conditioned texture generation, where late-stage over-control suppresses the synthesis of color and high-frequency texture. ControlNet-style scale control is closer in spirit, but it is still usually interpreted as a condition-adherence strength; directly applying a larger, smaller, or smoothly decayed control scale does not specify which denoising stage should receive strong geometric residuals for texture generation.

This distinction matters because the desired schedule for geometry-controlled texture generation is not necessarily monotonic. A monotonic decay or warm-up can reduce over-control in one part of the trajectory, but it cannot simultaneously avoid unstable early residuals, strengthen the middle stage where geometry refinement is most useful, and weaken late residuals that compete with color and high-frequency texture formation. ControlNet branch [3]. Thus, timestep-conditioned scheduling of adapter strength cannot be treated as a direct transfer of CFG scheduling or generic ControlNet scale scheduling; it requires a dedicated analysis of how geometric residual injection interacts with texture synthesislayer/time scheduling and geometry-conditioning schedules are established ideas. TCAS acts on the residual correction produced by GeoTex-Adapter from normal, depth, and foreground-mask conditions, with group-specific wrapper caps. In this 3D texturing setting, we measure how that intervention relates to generated-view agreement, reference-color errors, and baked appearance. SSI evaluates image style/content trade-offs. Our controlled question connects an implemented geometry-residual allocation to generated views and native-UV assets. The differences in task, operator and endpoints motivate this evaluation; they do not establish novelty of temporal or layer scheduling itself.

2.4. *Multi-view Adapter Interfaces*

MV-Adapter uses a condition guider for camera or geometry maps together with decoupled spatial, multi-view, and optional image cross-attention layers [23]. MVDiffusion uses correspondence-aware attention modules to connect pixels across views [24]. These interfaces differ from the GeoTex-Adapter residual branch studied here. We therefore report each auxiliary intervention within its own tested implementation and cohort, without pooling metrics or interpreting it as an architecture-matched replication of TCAS. The native MV-Adapter condition means are summarized in Table 6; its specified schedule primary and the external-input native-interface primary are reported with the original Fresh C timing primary in Table 7. The two separate bake primaries remain in Table 5; complete secondary families remain in Supplementary Sections S3–S5.

3. Control Formulation and Study Design

3.1. *Task Definition and Base Pipeline*

This paper builds on the MVPainter-style geometry-controlled multi-view texture generation framework [22]and studies how the We study inference-time strength of geometric condition injection affects the generated results. Figure 1 illustrates the overall pipeline: the adapter injects geometric residuals into UNet features, TCAS adjusts their strength across denoising stages, and the final geometry strength in an existing MVPainter-style multi-view outputs are projected and baked into a textured 3D mesh. Given a single reference image and the corresponding 3D geometry, the base pipeline first generates RGB texture images from six target views and then obtains a textured mesh through projection or texture baking. We do not modify the base UNet, VAE , or pretrained adapter weights texturing framework [22]. Instead, the adapter scaling factor is treated as the key inference-time control variableReference conditioning follows the base image path; normal, depth and mask conditions enter GeoTex-Adapter. TCAS sets the requested multiplier on the existing residual branch. The existing wrapper caps that value before multiplication and addition to frozen diffusion U-Net features (Figure 1). The base U-Net, VAE and adapter weights remain frozen. Fresh C evaluates six generated RGB views; separately identified bake cohorts assess actual UV-textured meshes from eleven unseen cameras. Baking is a downstream evaluation, not a module introduced by TCAS.

During multi-view diffusion, the 3D geometry is rendered into normal maps, depth maps, and foreground masks, denoted as:

$$\mathbf{G} = \{G_{\text{normal}}, G_{\text{depth}}, G_{\text{mask}}\}. \tag{1}$$

The geometric adapter maps these conditions to a residual correction term and injects it into the intermediate hidden states of the UNet. For the t-th denoising step inference step j with normalized execution progress p_j, the adapter injection can be written as:

$$\mathbf{h}'_{j,\ell} = \mathbf{h}_{j,\ell} + s_{\text{app}}(p_j, \ell)\, A^{\ell}_{\phi}(\mathbf{h}_{j,\ell}, \mathbf{G}), \tag{2}$$

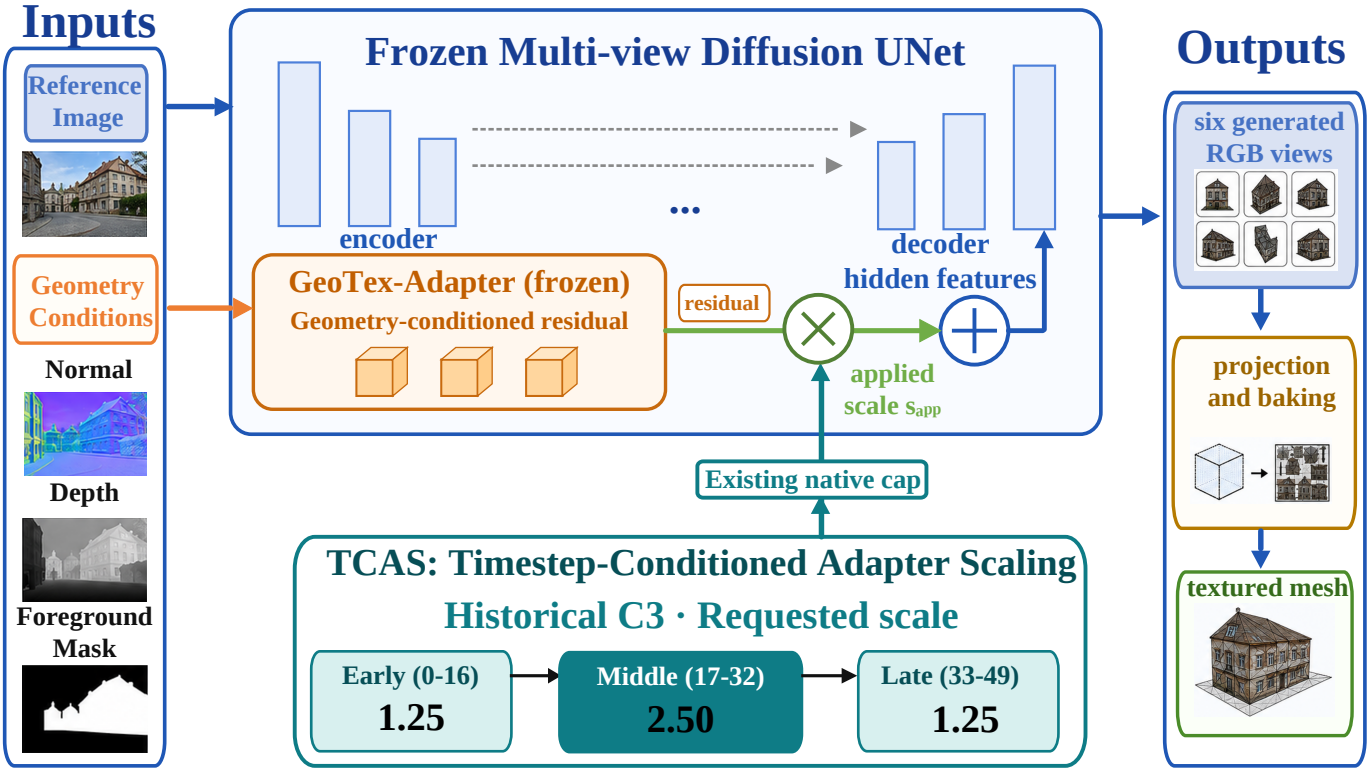


Fig. 1. Overview of the TCAS control interface in multi-view diffusion texturing. The frozen pretrained GeoTex-Adapter computes geometry-conditioned residual corrections from normal, depth and foreground-mask features, together with current hidden features. TCAS specifies the historical C3 requested scaling profile (1.25, 2.50, 1.25) over steps 0–16, 17–32 and 33–49. Existing native group-specific caps determine the applied scales before residual multiplication and addition to frozen U-Net features. In the evaluated Fresh C implementation, the deep, middle and shallow caps are 3.0, 3.5 and 0.8; shallow is clipped to 0.8 throughout. Six generated RGB views can subsequently be projected and baked onto a textured mesh. The image panels are illustrative and are not used as cohort evidence.

where $\mathbf{h}_\ell$ Here ℓ identifies a wrapper within one of the three injection groups; caps and requested profiles are assigned by group. The *requested scale* $s_{\text{req}}(p_j, \ell)$ is the value assigned by the profile; the *applied scale* is the value used by the wrapper,

$$s_{\text{app}}(p_j, \ell) = \min(s_{\text{req}}(p_j, \ell), c_\ell), \tag{3}$$

when the native cap c_ℓ is active, and otherwise equals the request. The resulting correction is $\mathbf{r}_{j,\ell} = s_{\text{app}}(p_j, \ell) A^\ell_\phi(\mathbf{h}_{j,\ell}, \mathbf{G})$; its *post-scale residual magnitude* is $m_{j,\ell} = \|\mathbf{r}_{j,\ell}\|_2$. Thus neither a requested nor an applied scale label alone determines the correction magnitude.

In the evaluated up-only Fresh C configuration, the GeoTex wrappers on UNet up blocks `up_0`, `up_1` and $\mathbf{h}'_\ell$ denote the hidden states before and after injection `up_2` are assigned to the deep, middle and shallow cap groups, respectively. These names identify the implementation's grouped injection points; they do not assert universal semantic roles for denoising stages. The corresponding Fresh C caps are 3.0, p_t denotes the sampling progress, and $s(p_t)$ is the adapter scaling strength at the current denoising stage. This formulation highlights a key difference from CFG-style guidance: changing $s(p_t)$ does not reweight two predicted scores, but changes the magnitude of a geometry-conditioned residual added to the feature stream. A larger s usually enhances geometric condition adherence. However, if strong residuals are continuously applied in the late denoising stage, they may suppress the synthesis of color, material, and high-frequency details by the base diffusion model. This paper therefore studies how to schedule s during denoising to balance shape consistency and texture detail preservation3.5, and 0.8. The shallow requests in all three C3 phases, 1.25/2.50/1.25, are applied as 0.80/0.80/0.80; deep and middle retain the requested profile. These caps belong to the existing wrapper and are not introduced by TCAS. Their values and group names do not establish causal semantic roles. GFL and GFH also clip at the shallow cap. These values describe the evaluated Fresh C wrapper; row-level cap semantics for the historical 24-object probe are not recoverable. TCAS scales a geometry residual added to feature activations; it does not reweight conditional and unconditional score predictions as classifier-free guidance does.

3.2. *Image-Space Metrics and Texture-Variation Diagnostics*

The influence of adapter scaling on generated results cannot be evaluated by a single structural metric. We divide evaluation metrics into three groups. The first group consists of shape and structure metrics, including FG-SSIM, Edge-SSIM, and Crop-SSIM, which measure consistency with ground-truth renderings in the foreground, edge regions, and local foreground crops based on the structural similarity index [39]. The second group contains signal fidelity and perceptual similarity metrics, including PSNR and the learned perceptual image patch similarity (LPIPS) [40], which reflect pixel-level error and differences in deep feature space. The third group consists of texture diagnostic metrics, including Laplacian variance, endpoints describe different outcomes. *Image-space agreement* uses target-view PSNR, LPIPS, and SSIM on the named regions; it does not measure final 3D material quality. *Reference-color agreement* uses measures such

as CIEDE2000 against the corresponding reference. *Texture variation* uses RGB standard deviation, and Laplacian variance, gradient magnitude, which are used to detect weakened color variation, surface smoothing, and high-frequency detail loss.

Taking Laplacian variance as an example, the foreground texture response is defined as: where Lap denotes the Laplacian filter, G_{mask} denotes the foreground mask, and Var denotes spatial variance. A larger M_{tex} usually corresponds to richer local high-frequency variation. If structural metrics improve while Laplacian variance, RGB standard deviation, or gradient magnitude decreases, the improvement may mainly come from contour smoothing or boundary convergence rather than true texture quality improvement. energy; these can increase with detail or artifacts and are not fidelity measures. *Subjective preference* refers only to the submitted forced-choice study. *Baked appearance* refers to rendered meshes after texture projection or baking, whereas *PBR/material fidelity* additionally depends on the material and illumination model. These endpoints are not interchangeable: FG-SSIM, Edge-SSIM, and Crop-SSIM quantify structural similarity on named regions, PSNR measures pixel error, and LPIPS compares learned image features [39, 40]. Texture-variation statistics describe local image properties; neither a higher value nor a lower one alone establishes fidelity or reference preservation.

3.3. TCAS Control Semantics and Comparison Design

The diffusion denoising process is stage-dependent: early steps mainly establish the global layout and coarse contours; middle steps refine geometric structure and cross-view alignment; late steps are more responsible for color, material, and local texture synthesis. Based on this observation, we propose Timestep-Conditioned Adapter Scaling (TCAS), which

TCAS replaces a globally fixed requested adapter scale with a stage-dependent scale function. TCAS is not intended as a generic smooth guidance schedule. Its low-high-low form encodes the adapter-specific hypothesis that geometry residuals should be strongest only after coarse structure becomes stable and before late texture synthesis dominatespiecewise profile over normalized sampling progress. The early/middle/late partition is an operational description, not a universal claim about the function of every denoising stage. C3's low–high–low profile is the recorded development choice evaluated in this study; it is not presented as an independently derived optimum.

To avoid ambiguity caused by different raw timestep directions in different schedulers, we use sampling progress p to represent the denoising process, where $p = 0$ denotes $p_j = j/(T-1)$ denotes execution progress for inference step $j \in \{0, \dots, T-1\}$. Thus $p = 0$ is the first update from the start from noise and $p = 1$ denotes the end of sampling close to the final image state. TCAS divides denoising into early, middle, and late stages: initial noisy latent and $p = 1$ is the last denoising update before decoding. For $T = 50$, TCAS divides execution into steps 0–16, 17–32, and 33–49 (17/16/17 steps):

$$s_{\text{req}}(p) = \begin{cases} s_e, & 0 \le p < 1/3 \\ s_m, & 1/3 \le p < 2/3 \\ s_l, & 2/3 \le p \le 1 \end{cases}. \tag{4}$$

We call this variation across the 50 inference steps the *temporal allocation*. A *layer allocation* varies the requested scale across the three wrapper groups at a fixed step. For a schedule with applied values $s_{\text{app}}(j, \ell)$, its per-group applied-scale sum is $B_\ell = \sum_j s_{\text{app}}(j, \ell)$; the unweighted sum $B_{\text{nom}} = \sum_\ell B_\ell$ is a *nominal applied-scale budget*. It does not weight scales by adapter activations. We log post-scale correction summaries, including wrapper-level L_2 norm and mean absolute entry. The three wrapper groups operate on feature maps with different spatial dimensions, and the traces do not retain channel counts; raw L_2 values are therefore not used for cross-layer dose comparisons. The per-entry mean absolute correction reported in Supplementary Section S1 is a dimension-normalized descriptive summary, not an equal-dose intervention. Correction magnitudes also depend on the current hidden state and earlier denoising updates, so equal nominal budgets do not imply equal realized corrections.

The selected C3 configuration used in this paper is:

$$(s_e, s_m, s_l) = (1.25, 2.50, 1.25). \tag{5}$$

This design follows three principles. First, the early stage uses a conservative scale to avoid disrupting global structure formation with strong geometric residuals. Second, the middle stage uses a higher scale to strengthen geometric constraints from normals, depth maps, and foreground contours. Third, the late stage reduces the scale again to prevent geometric residuals from suppressing color variation, material detail, and high-frequency texture synthesis. This non-monotonic schedule is therefore different from simply decaying or increasing the adapter strength across all timesteps. TCAS introduces no new The low–high–low profile is a concrete inference-time intervention motivated by the submitted development analysis. Its stage interpretation is a task-specific hypothesis: the tested early, middle, and late allocations may produce different responses. Fresh C evaluates the recorded profile against matched controls but does not prove a universal function for any stage. TCAS adds no trained parameters and does not modify the base model base-model or adapter weights; it only replaces the scaling factor during inference, so its computational overhead is negligiblereplaces the requested scale during inference.

What C3 changes and what its comparisons identify. C3 requests $(1.25, 2.50, 1.25)$ for each of the 50 steps across the three layer groups before native caps. In Fresh C, the applied deep and middle profiles retain those values; the shallow profile is clipped to 0.8 throughout. GFL requests 1.25 at every step and group, so its deep and middle values are 1.25 throughout and its shallow value is also clipped to 0.8. Their per-group applied-scale sums are therefore 82.5 versus 62.5 in deep and middle, and 40 for both in shallow. The low–high–low form is not assumed but derived from a measurable Correction–Alignment–Interference (CAI) stage-utility model; we state the selection rule formally. Let $Q(s_e, s_m, s_l)$ denote a fidelity objective (PSNR here), let L denote the fixed-low schedule, let R denote the texture-deviation risk, given by the artifact-sensitive foreground texture diagnostics of Section 3.2 (Laplacian variance, RGB standard deviation, and gradient magnitude), and let $z_k \in \{0, 1\}$ indicate whether the high scale is assigned to stage $k \in \{e, m, l\}$. Risk is compared by a non-worsening (Pareto) rule rather than a weighted scalarization: one assignment has lower risk than another only if it is no worse on every diagnostic and strictly better on at least one, so no artificial weights are introduced.

Suppose (i) the stage-utility model $Q(s_e, s_m, s_l) = Q(L) + \sum_k U_k z_k$, where $U_k = Q(\text{high scale only in } k) - Q(L)$ is the measured finite-difference utility of stage k, and (ii) per-stage guardrails under which stage k is admissible for the high scale only if its intervention does not reduce FG-SSIM by more than τ nor raise the texture-deviation risk beyond a factor ρ relative to L. Selection proceeds in two stages. First, maximize fidelity over the admissible schedules $\mathcal{F}$: the fidelity-optimal assignment is $z_k^{\text{fid}} = \mathbf{1}[U_k >$

0 and k admissible], with value $Q_{\max} = \max_{z\in\mathcal{F}} Q(z)$. Second, restrict attention to the ϵ-equivalent set $\mathcal{F}_\epsilon = \{z \in \mathcal{F} : Q(z) \geq Q_{\max} - \epsilon\}$ for a small fidelity-equivalence margin $\epsilon > 0$, and select $z^\star = \arg\min_{z\in\mathcal{F}_\epsilon} R(z)$. Then: an inadmissible stage is fixed at the conservative scale; any admissible stage whose estimated utility alone exceeds ϵ must retain the high scale ; and a stage whose estimated utility satisfies $0 < U_k < \epsilon$, with a structurally insignificant change, is fidelity-equivalent under either assignment, so the risk-dominant conservative scale is selected. Measured confidence intervals serve as auxiliary uncertainty information and do not enter the membership condition of $\mathcal{F}_\epsilon$. C3–GFL contrast changes the middle-third deep/middle scale magnitude as well as its temporal allocation; it is not a timing-only comparison.

Proof. Under (i), for any $z \in \mathcal{F}$, $Q(z) = Q_{\max} - \sum_{k:z_k^{\text{fid}}=1} U_k\,(1 - z_k)$, so $z \in \mathcal{F}_\epsilon$ exactly when the total positive utility dropped by reassigning high-scale stages to the conservative scale does not exceed ϵ. Condition (ii) fixes $z_k = 0$ for inadmissible stages and leaves the admissible stages independent, so each stage can be examined separately. A stage with $U_k > \epsilon$ cannot be dropped from the fidelity optimum without leaving $\mathcal{F}_\epsilon$, hence $z_k^\star = 1$. A stage with estimated utility $0 < U_k < \epsilon$ contributes at most ϵ when dropped, so both assignments lie in $\mathcal{F}_\epsilon$; within $\mathcal{F}_\epsilon$ the second selection stage minimizes texture-deviation risk, and for such a stage the structural change is insignificant while the conservative scale strictly reduces the risk, so $z_k^\star = 0$. □ To identify a temporal-placement contrast, controls hold the per-layer applied values and nominal per-group sums fixed while moving the same values between time windows. A scale-strength contrast instead changes amplitude while preserving layer and temporal allocation. Matching requested values or applied-scale sums does not match the evolving residual corrections; raw L_2 values are also dimension-sensitive across wrapper groups and cannot establish cross-layer dose matching. The 300-object allocation comparison matches applied sums in each layer group and therefore estimates an allocation effect at a fixed scalar budget, not a residual-dose-independent mechanism. The historical-request sensitivity panel is reported separately.

The utilities are measured finite differences on a 24-object stage ablation of the strong-residual, per-layer-capped adapter instance (regime (b) of Section *[removed reference:* `subsec:adapterdep`*]*), where per-stage effects are most visible: $U_e = -0.788$ dB (95% CI $[-1.420, -0.156]$) with a guardrail violation (FG-SSIM -0.105, CI $[-0.141, -0.069]$; foreground Laplacian variance 3.85× fixed-low), $U_m = +0.574$ dB $[0.391, 0.756]$, and $U_l = +0.077$ dB $[0.006, 0.149]$ with a structurally insignificant FG-SSIM change ($+0.001$, CI $[-0.002, 0.003]$). Proposition *[removed reference:* `prop:cai`*]* selects exactly the conservative–high–conservative assignment

Development origin and scope of C3. The submitted study records C3 : the early stage is inadmissible, the middle stage must retain the high scale because its utility alone (+0.574 dB) exceeds any meaningful equivalence margin such as $\epsilon = 0.1$ dB, and the late stage has estimated utility $U_l = 0.077 < \epsilon = 0.1$ dB while its FG-SSIM change is statistically insignificant, so removing the late high-scale intervention remains within the ϵ-equivalent fidelity set and reduces texture risk; the risk-dominant conservative scale is therefore selected. The same conservative–high–conservative assignment is independently selected by the main adapter's 17-variant probe (Table *[removed reference:* `tab:probe`*]*); the utilities above explain, rather than determine, that selection, and Remark 2 delimits the conditional scope of this explanation.

Additivity is an approximation: stage effects interact, and the complete $=(1.25, 2.50, 1.25)$ as the selected profile from a 24-object development probe containing 17 uniform, layer-wise, and time-conditioned variants. The available historical records do not preserve the row-level outputs needed to reconstruct that ranking. We therefore treat C3 schedule is therefore validated directly (Secs. *[removed reference:* `subsec:c3`*]*–*[removed reference:* `subsec:adapterdep`*]*). Changing the adapter, its residual magnitude, or the scale semantics changes the measured U_k and requires re-estimation. Proposition *[removed reference:* `prop:cai`*]* thus provides a conditional decision rule within the measured adapter regime—explaining why the mid-stage assignment follows from data rather than from a hand-tuned rule—and not a universal guarantee.

GeoTex-Adapter, TCAS, and FAC correspond to three complementary dimensions of adapter control. GeoTex-Adapter determines what geometric residual correction should be injected. TCAS determines when the residual should be strengthened or weakened along the denoising-time dimension. FAC further modulates the residual in a fine-grained manner along spatial positions and frequency componentsas the recorded operating point and make no optimality or uniqueness claim based on the original probe. Fresh C evaluates it without further schedule search.

3.4. Geometry Adapter and Inference Interface

The pretrained GeoTex-Adapter is used as a fixed geometric residual backbone. It receives a 5-channel geometric input consisting of a 3-channel receives a five-channel input comprising a three-channel normal map, a 1-channel one-channel depth map, and a 1-channel one-channel foreground mask. A Its multi-scale geometric encoder produces a feature pyramid, and residual corrections encoder produces residual features that are injected into multiple resolution levels of the the frozen UNet. TCAS acts on the global temporal scale of these residuals to control the influence of geometric conditions at different denoising stages without changing the adapterarchitecture.

To further examine whether stage-wise adapter control can be extended to learned adaptive correction, we also conduct experiments with Full Adaptive Correction (FAC). FAC contains three lightweight modules: LTAG learns timestep-dependent gating, GSG generates spatial gates from geometric features, and FSC performs frequency-selective correction on adapter residuals. Unlike fixed TCAS, FAC finely modulates adapter residuals in the temporal, spatial, and frequency domains. In the extension experiments, FAC uses TCAS as the base temporal schedule and further adds GSG and FSC for spatial and frequency correction.

It should be emphasized that FAC requires learning a small number of additional parameters. Therefore, TCAS,which requires no training , remains the main method in this paper. FAC is used only as a learned adaptive extension; Section *[removed reference:* `subsec:facexp`*]* reports a controlled re-examination that delimits when learned modulation can be expected to helprequested multiplier along denoising progress; it does not retrain the adapter, the UNet, or the VAE and introduces no learned parameters.

Visual comparisons use recorded outputs with shared object framing and crops. They illustrate task behavior and appearance limitations; quantitative conclusions use the stated object-level metrics.

3.5. Study Cohorts, Sampling and Statistical Units

Table 1 summarizes the cohorts and protocols. The principal evaluation cohort, denoted Fresh C, contains 300 object identities disjoint from the historical schedule-selection/evaluation pool, the recorded 1,118-object training list and the earlier clean-v2 evaluation list. Its primary four-condition comparison uses one frozen retrained GeoTex checkpoint, shared inputs and paired object seeds. This study does not reproduce the historical adapter's reported +0.96 dB estimate; the original main checkpoint's exact identity remains unavailable. A separate historical-request retraining is reported as protocol sensitivity.

Experiments are conducted using an MVPainter-style geometry-controlled multi-view diffusion pipeline. Each objectcontains a single reference image and corresponding 3D geometry. During preprocessing, the 3D geometry is rendered into target-view normal maps, depth maps, foreground masks, and ground-truth RGB images. The model generates six target views at a time, and all comparisons with ground-truth renderings are performed under unified views, resolution, and foreground masks

Within each comparison, object inputs, targets, initialization and endpoint code are shared. Fresh C uses seed 42 + `object_idx`, held constant across conditions for that object. Its capped residual groups have deep/middle/shallow limits of 3.0/3.5/0.8. MV-Adapter uses a different feature injection operator and numerical scale range; its native target cameras and actual default-path identity are checked separately. Scale labels are not common physical doses across interfaces.

Except for the FAC extension experiment, all scaling experiments keep the object list, reference image, target views, random seed, base UNet, VAE, adapter weights, and number of denoising steps fixed; only the adapter scale or schedule is changed. The number of sampling steps is 50. C3 is selected using only a 24-object probe set; the same schedule is then frozen and applied to the 300-object evaluation pool, where no parameter search is performed. This probe-to-pool protocol isolates the effect of adapter scaling from changes in model architecture, training data, Fresh C's controlled sampler differs from the complete base pipeline, which uses online image features, classifier-free guidance and 512-pixel views. Constant 1.0 under the GeoTex caps also differs from the unset-scale residual branch. We therefore name these controls by their executed configuration. The source records and sampling budget, or evaluation-set tuning. /default-path diagnostics are in Supplementary Sections S1 and S6.

Accordingly, the comparisons in this paper are comparisons among inference-time adapter scaling strategies under the same base pipeline, rather than a system-level benchmark among different texture generation architectures. Uniform adapter scales serve as the primarydeployment baselines because they correspond to the most direct way of choosing a single fixed adapter strength at inference time. The probe variants further test whether the trade-off can be resolved by simple alternatives such as globally high scales, layer-wise redistribution, or moving high adapter strength to early, late, or multiple denoising stages. TCAS changes only how this strength is scheduled across denoising stagesFG-PSNR, LPIPS, SSIM and Edge-SSIM measure agreement on the named image regions. Baked evaluation renders actual exported meshes; unlit base color does not establish PBR or material fidelity. Fresh C image endpoints are evaluated once on each frozen 3×2 six-view composite. Native-interface and baked endpoints average their recorded per-view scores within object, with GSO sampling seeds also averaged within object. Paired intervals resample objects and are pointwise 95% intervals. Preserved primary, secondary and exploratory families use their stated Holm adjustments; the distinct families do not imply one overall experiment-wide error guarantee. Non-detection is not equivalence or non-inferiority. Complete comparison rules are supplied in Supplementary Sections S2–S5.

Reproducible analysis and presentation. Statistical analyses use NumPy/SciPy with the specified object resampling, seeds and original correction families; plots use Matplotlib. The accompanying scripts and source records document the calculations and inputs for numerical verification.

We use four evaluation sets: (1) a 50-object scale-sweep set for analyzing the overall trend of structural metrics under 13 uniform scales; (2) a 26-object texture audit set for comparing texture degradation under $s = 1.25$ and $s = 2.50$; (3) a 24-object probe set for evaluating 17 uniform, layer-wise, and timestep-conditioned scheduling variants and selecting

4. Experimental Results

4.1. Competitive Controls and the Historical C3 Profile

The internally specified Fresh C primary is C3 ; and (4) a 300-object evaluation pool, comprising the 24-object probe and a 276-object held-out subset (detailed below), for testing the selected minus Fixed Low (GFL) FG-PSNR. The paired gain is +0.52 dB (95% interval [+0.40, +0.64]), with 206/300 objects favorable. All conditions use the same frozen retrained checkpoint and object inputs. This gain combines increased strength and temporal allocation: C3schedule, uniform baselines, and the FAC extension without any further schedule search. 's deep/middle applied sums are 82.5 each, versus 62.5 for GFL, while both have shallow sum 40.

Evaluation metrics are divided into three categories: shape and structure metrics (

Competitive controls change the recommendation. Fixed High has better means than C3 on all four endpoints: C3 minus Fixed High is −0.748 dB FG-PSNR (pointwise interval [−0.885, −0.609]), with FG-SSIM difference −0.04553 ([−0.04937, −0.04167]). Generic Linear ramps deep/middle requests from 1.25 to 2.50 and shallow requests from 0.50 to 0.75 over the fifty steps. Its applied sums are (93.75,93.75,Edge-SSIM, and Crop-SSIM), signal and perceptual metrics (PSNR and LPIPS), and texture diagnostic metrics (Laplacian variance,RGB standard deviation, and gradient magnitude). All metrics are first computed over six target views and then averaged over views and objects. For texture-related conclusions, we report means, 31.25), so it changes layer allocation and total budget as well as timing. It also has better means on all four; C3 minus Generic Linear is −0.593 dB ([−0.837, −0.351]). Its positive median of +0.074 dB and 155/300 favorable objects show why mean and object-level win rates, and significance tests to avoid judging generation quality based on a single structural scoredistributions must both be considered. Complete original paired endpoints and their comparison families are in Supplementary Section S2.

The FAC extension uses the same object list and evaluation protocol as the 300-object evaluation pool, and its lightweight modules are trained only on the 1, 706-object training pool for the strong-residual adapter instance, which is disjoint from all evaluation objects. TCAS is used as the main baseline, while LTAG, LTAG+GSG, and Full FAC are evaluated as ablation variants under the same paired protocol. The base model and original adapter remain frozen, and only lightweight adaptive correction parameters are learned.

All objects are drawn from the Objaverse 3D asset collection: ground-truth multi-view images are rendered offline with Blender (Cycles engine, 17 views at 512×512, white background), and six target views (front, front-right, right, back, left,front-left) are used throughout the paper. The main adapter's training pool contains 1,118 rendered objects, while the strong-residual v2 instance used in Section *[removed reference:* `subsec:adapterdep`*]* and the FAC experiments is trained on a separate 1

4.2. Temporal Allocation at Matched Applied-Scale Budgets

Early-, 706-object pool from the same source. Both training pools are disjoint from the evaluation pool; we verified that each trainingmiddle-, and late-high move the same sixteen high steps to updates 0–15, 17–32, and 34–49. Deep/evaluation identifier intersection is zero. Within the 300-object evaluation pool, the 24-object probe set corresponds to the first 24 identifiers

Table 1. Studies and evidence scope. Cohorts may overlap; views and seeds remain nested within object.

Study / objects	Interface and sampling	Endpoint	Selection and scope
Fresh C / 300	Capped GeoTex residual; 256 px; 50 Euler steps; conditional-only	Six target-view FG metrics	UID-disjoint cohort; C3 primary; linear and matched-budget addenda
GeoTex bake / 10	Controlled generator; native UVs; six bake views	Eleven unseen unlit views	Compatibility-selected pilot; Late–Mean primary; two later controls exploratory
MV-Adapter / 50	Native condition-guider; 512 px; 50 steps; CFG 3	Six native-camera views	Input-selected; thirteen legal white targets retained
GSO / 100	Native MV-Adapter; 512 px; 50 steps; CFG 3; three seeds	All-six PSNR primary; non-reference LPIPS secondary	Source-ordered convenience cohort; external sensitivity
GSO bake / 20	Fixed GSO inputs; native UVs; three seeds	Eleven unseen unlit views	GT-only recalibration after initial outcomes; full cohort retained
Reference protocol / 50	New frozen GeoTex; native references; 512 px; CFG 2	Five nonreference views	Previously studied inputs; protocol/checkpoint sensitivity
Historical request / 276 slots	Retrained GeoTex; 1,039 training UIDs; one unknown replacement	Six-view composite; full-frame and FG	Exposed technical sensitivity; exact held-out UID coverage uncertified
MVDiffusion / 75	Partial correspondence-block scaling; depth unscaled	Recorded six-view endpoints	Retrospective third-interface panel; not exact GeoTex operator transfer

Table 2. Fresh C competitive-control means on the same 300 objects.

Control	Deep/middle sum	Shallow sum	FG-PSNR↑	FG-LPIPS↓	FG-SSIM↑	Edge-SSIM↑
Fixed Low	62.5	40	11.31	0.2051	0.3500	0.4728
Fixed High	125.0	40	12.58	0.1891	0.4201	0.4961
Fixed Mean	82.5	40	11.96	0.1978	0.3864	0.4843
Early-high	82.5	40	11.33	0.2045	0.3696	0.4765
C3 / Middle-high	82.5	40	11.83	0.1998	0.3746	0.4773
Late-high	82.5	40	12.39	0.1951	0.3788	0.4914
Generic Linear†	93.75	31.25	12.42	0.1862	0.4026	0.4880

Equal-budget rows (Fixed Mean, Early-high, C3/Middle-high and Late-high) share per-group sums of 82.5 for deep/middle and 40 for shallow. Generic Linear† is a same-cohort addendum with different sums (93.75, 31.25); it is not an equal-budget strategy.

(obj_{0000}–obj_{0023}) and the remaining 276 identifiers (obj_{0024}–obj_{0299}) form a holdout that is strictly disjoint from the probe; Section *[removed reference:* `subsec:adapterdep`*]* reports transfer statistics on this disjoint holdout for the strong-residual,per-layer-capped adapter instance, in addition to the full-pool tables. The 50-object scale-sweep set comprises the first 50 objects of the pool (obj_{0000}–obj_{0049}), and the 26-object audit set is a purposive subset of these 50 objects, selected to span the observed range of per-object ΔFG-SSIM behavior for manual inspection; all evaluation sets are subsets of the 300-object pool, middle low and high are 1.25 and no training object appears in any of them2.50; shallow stays at its 0.8 cap. Fixed Mean uses 1.65 before the cap. Each allocation therefore has per-group sums $(82.5, 82.5, 40)$. All six conditions share the object inputs and seed, giving 1,800 paired object-condition rows.

The equal-budget addendum's internally specified primary question is Middle-high minus the objectwise average of Early-high and Late-high FG-PSNR. Its mean is -0.0339 dB (10,000-resample pointwise object-bootstrap interval $[-0.1172, +0.0514]$; two-sided 20,000-draw sign-flip $p = 0.4324$), with 119/300 objects positive. The median is -0.2296 dB. This result does not support a middle-window advantage and does not establish equivalence. It was specified before the addendum's new allocations, after earlier Fresh C outcomes were known; it is not an independent prospective confirmation. The primary, twelve-test secondary family and later descriptive all-pair analysis are distinguished in Supplementary Section S2.

The base pipeline is an MVPainter-style geometry-controlled multi-view diffusion model built on a joint six-view latent UNet, loaded from the local `MVPainter_Pipeline` snapshot under `checkpoints/hf_repo` with diffusers 0.20.0, and used frozen together with its VAE. The pipeline is reference-image-conditioned; no text prompts are used. The GeoTex-Adapter takes a five-channel geometric input (3-channel normal map, 1-channel depth map, 1-channel foreground mask) , produces a multi-scale residual pyramid, and injects corrections into multiple UNet resolution levels; for the adapter studied in the main tables, the requested scale is applied multiplicatively at every injected layer. Inference uses the Euler discrete scheduler with 50 denoising steps and a fixed random seed (42)shared by all schedules within an experiment, so all variants share the same initial latents; the model generates six views per object.

FG-SSIM, Edge-SSIM (edge masks derived from depth/normal discontinuities)near-zero middle-versus-average primary contrast should not be interpreted as insensitivity to temporal placement: the retrospective Middle-high contrasts against Early-high and Late-high have opposite signs and largely cancel in that average. Later descriptive all-pair comparisons place Middle-high (C3) above Early-high by $+0.496$ dB and below Late-high by -0.564 dB in FG-PSNR; both have Holm-adjusted $p = 0.00059$ in the separate retrospective 59-test family (their reported intervals are pointwise). The corresponding Middle–Early and Middle–Late FG-LPIPS differences are -0.00472 and $+0.00466$ (both Holm-adjusted $p = 0.00059$), PSNR, and LPIPS (AlexNet backbone) are computed on foreground-masked regions after background normalization; the texture diagnostics (foreground Laplacian variance, RGB standard deviation, gradient magnitude) are computed on foreground pixels and compared against the same ground-truth renderings under identical views and masks. The

main tables use GeoTex-Adapter `geotex_refattn_v1/geotex_step_0002000.pt` trained on the 1, 118-object pool; the strong-residual analyses use `mvpoutput/geotex_v2/checkpoints/geotex_v2_ema_final.pt` (MD5 `a74cc1d16cb473a95022bdcbc58e1b22`) trained on the 1, 706-object pool. The checkpoint files and evaluation code will be made available upon reasonable request, and a consolidated reproducibility table (data subsets, checkpoint identifiers, scheduler, seed, and metric implementations) is provided in the supplementary materialwith the same ordering. Middle–Fixed Applied Mean FG-PSNR is −0.133 dB (pointwise interval [−0.170, −0.096]; its raw $p = 0.00001$ is exempt from that family). These contrasts indicate within-cohort sensitivity to temporal placement at matched applied sums, but do not replace the specified timing primary, establish a residual-dose-independent mechanism, or validate a universally preferred late schedule. Full contrasts and family details are in Supplementary Section S2. Within these six conditions Fixed High wins FG-PSNR on 172 objects, Late-high on 64, and C3 on none; this is an endpoint-specific count, not a universal quality ranking.

Object-level paired comparisons report two-sided paired tests and 95% confidence intervals constructed by percentile bootstrap over objectswith 10, 000 resamples. Where a mean is accompanied by ±, the quantity is the standard deviation across objects . Preference-study intervals add a participant-level bootstrap level (Section 4.1)A descriptive four-endpoint comparison complements these winner counts. On 270/300 objects, at least one of the other five controls weakly dominates C3 on all four recorded composite endpoints and improves at least one. Fixed High alone does so on 173 objects, but C3 dominates it on 16; their remaining 111 objects have competing endpoints. This post-inspection, point-estimate analysis is not statistical separation on each object or a ranking of unmeasured material quality. Full counts are in Supplementary Section S2.

Table *[removed reference:* `tab:sweep`*]* reports the results of a 13-scale uniform adapter sweep on 50 objects.

As the scale increases from $s = 0.75$ to $s = 2.50$, ΔThe equal-budget comparison identifies an allocation total effect. Evolving residual amplitudes remain trajectory-dependent, preventing a residual-dose-independent mechanism interpretation. Late-high is not universally preferred: Fixed Mean has higher FG-SSIMrises from −0.013 to +0.145, and the number of objects with positive improvement increases from 20/50 to 44, and Generic Linear is a strong unmatched-budget practical control. The historical LLH layer/50. This indicates that stronger adapter residuals indeed improve the model's adherence to geometric conditions such as normals, depth maps, and foreground contours. Figure *[removed reference:* `fig:curve`*]* visualizes this trend: FG-SSIM keeps increasing with adapter scale, while Crop-SSIM saturates and slightly decreases at high scalestime addendum and complete object distributions remain in Supplementary Section S2.

Overall, the FG-SSIM improvement is stable, but the marginal benefit becomes smaller in the high-scale region. For example, increasing the scale from $s = 2.25$ to $s = 2.50$ yields only a limited FG-SSIM gain, while Crop-SSIM already decreases. This suggests that the nature of the high-scale benefit may change: the gain comes more from contour smoothing and foreground alignment than from preservation of local texture details. Figure *[removed reference:* `fig:heatmap`*]* showsthe per-object ΔFG-SSIM heatmap, confirming that most objects gain higher structural scores as the scale increases, but with clear object-level differences.

Table 3. Selected paired color-diagnostic contrasts.

Contrast	ΔE_{00}	95% CI	Favorable
GFL – No Adapter	−19.654	[−26.370, −13.248]	24/26
LLH – GFL	+3.583	[+0.462, +6.712]	–

Failure-enriched $N = 26$; negative differences favor the first condition. Favorable is an object-direction count, not prevalence.

4.3. Appearance and Color Failure Analysis

To determine whether the FG-SSIM gain at high scales reflects better visual quality, we further conduct a texture audit on The color diagnosis is a failure-enriched set of six earlier cases and 20 Fresh C panels, paired on the same inputs, checkpoint and runner across six target views; it is not a random sample, a new holdout or a prevalence estimate. As Table 3 shows, GFL lowered reference-relative CIEDE2000 in 24/26 objects, comparing texture metrics and foreground shape changes for $s = 2.50$ relative to $s = 1.25$. Figure *[removed reference:* `fig:qualitative`*]* should be read as a local texture comparison rather than as a claim that any method fully matches the ground truth: the key evidence is the progressive loss of color variation , material marks, and high-frequency patterns inside the marked regions as the scale increases. The conservative scale $s = 1.25$ better preserves color variation and local texture, while $s = 2.50$ more easily produces uniform and smooth surfaceswhile pink or purple shifts persisted in two legacy examples. LLH reduced local variation while increasing mean color error, so lower variation did not imply closer appearance. The shifts were present before UV projection or baking; full measurements, VAE/runtime isolates and failed correction candidates are in Supplementary Section S6 and establish no complete cause or repair.

The results show that 24 out of 26 objects (92%) exhibit a pattern of "texture loss without true shape gain." Only two objects show comparable texture quality, and no object shows a clear texture improvement from the high scale. Here, "true shape gain" is not equivalent to an increase in FG-SSIM. It refers to whether the high scale brings substantive improvement in effective foreground area, contour structure, or geometric detail compared with the conservative scale. If the metric improvement mainly comes from smoothed foreground boundaries or averaged contour errors without visible geometric improvement, it is categorized as having no true shape gain. Therefore, although a high adapter scale can improve structural metrics, such improvement does not necessarily indicate better visual shape or texture quality. Figure *[removed reference:* `fig:audit`*]* further quantifies this texture loss.

For multi-view texture generation, overly strong geometric control can suppress the color and detail synthesis capability of the base diffusion model during late denoising steps, making surfaces more uniform. Therefore, scale selection should not rely only on standard metrics such as FG-SSIM or PSNR, but should also incorporate texture-sensitive diagnostics.

Earlier bake results and calibration checks are reported separately in Supplementary Section S5 and are not pooled with the ten-object cohort. The GeoTex residual branch with an unset scale differs from an explicit Constant 1.0 request under the wrapper caps; we label the latter by its executed setting.

Based on the texture audit , we do not pursue higher adapter scales throughout the entire denoising process. Instead, we further explore how different scaling strategies balance shape gain and texture loss

Full-object evidence and its selection. The complete views in Figure 2 and the color failure in Figure 4 use a shared full-

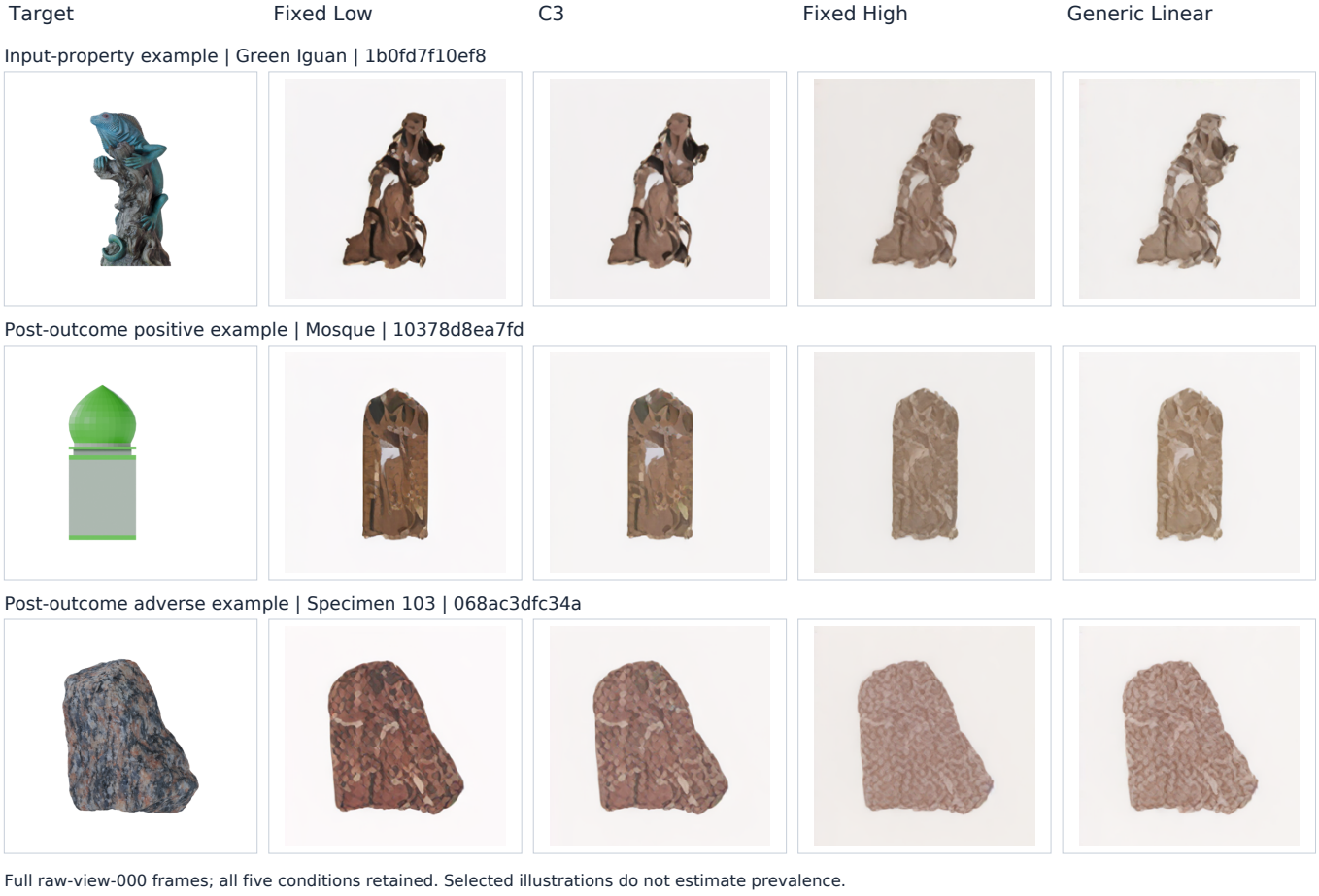


Fig. 2. Fresh C, camera 000. Every row shows Target, Fixed Low, C3, Fixed High and Generic Linear in the same full-frame view. Targets are source RGB conditioning renders, not measured ground-truth albedo. Green Iguan is a colorful, textured organic object selected from source properties. Mosque (+1.736 dB) and Specimen 103 (−0.468 dB) are post-outcome selected positive and adverse C3–Fixed Low FG-PSNR examples. These selected rows illustrate endpoint heterogeneity; they do not estimate prevalence or establish a cohort-wide advantage. Complete source identity and credits are in Supplementary Section S9 and the visual selection manifest.

frame display for every condition within each object; the existing color diagnostic retains its identical pixel crop. Favorable/adverse C3–GFL examples are retained in Supplementary Section S9. We construct 17 adapter scaling variants and evaluate them on the 24-object probe set. To avoid turning the 300-object evaluation pool into a hyperparameter search, the schedule selected in this probe stage is frozen before large-scale evaluation. These variants include three categories: uniform scaling baselines, layer-wise scaling strategies, and timestep-conditioned scaling strategies. Uniform scaling tests the effect of globally strengthening adapter influence and serves as the closest counterpart to using a single ControlNet-style condition scale. Layer-wise scaling examines whether redistributing the same geometric intervention across UNet levels is sufficient. Timestep scaling tests stage placement directly, including variants that put high adapter strength in early, middle, late, or broader temporal regions. This design is intended to rule out the simple explanation that any guidance-like schedule or any stronger control scale would solve the problemOutcome-selected displays illustrate behavior, not prevalence. The historical texture audit does not preserve a traceable denominator or validated texture-loss labels; we therefore do not infer color-failure prevalence or texture loss from it. Variation statistics alone cannot establish fidelity.

4.4. Actual Baked Appearance: Local Benefit and External Limits

4.4.1. Native-UV protocol and primary result

We selected ten previously unbaked Fresh C objects in UID order using input availability, native UV/material compatibility, triangle count and camera-mask alignment. Only ten of the 300 screened inputs passed these criteria before the cutoff. Their generated-view outputs were known, but their baked outcomes had not been inspected during selection. This compatibility-selected pilot is not a representative sample; five earlier bake objects are reported separately.

All variants are evaluated using ΔFG-SSIM for shape gain and ΔLap Var for texture detail change. Delta texture metrics are computed relative to the corresponding ground-truth foreground renderings under the same views and masks. Importantly, the objective of this probe experiment is not to find the single highest structural score, but to identify a stable compromise strategy that avoids the trap of "higher structural metrics but flattened texture. " This criterion is crucial: if a schedule improves FG-SSIM by applying strong geometry residuals late but causes a clear drop in texture response, then it does not solve the adapter-specific conflict even though it appears favorable under a structure-only metricSeven conditions share each object's source mesh, reference, seed, six projection cameras, masks, 256-pixel

Table 4. Ten-object native-UV bake pilot: mean scores over eleven unseen views.

Condition	FG-PSNR	FG-LPIPS	FG-SSIM	CIEDE2000
GT-256 reference bake	22.38	0.0590	0.7683	4.44
Native residual†	11.13	0.1372	0.3376	26.47
No Adapter†	5.92	0.1600	0.3829	40.92
Constant 1.0 (capped)	11.93	0.1354	0.3504	24.18
Fixed Low	12.48	0.1317	0.3835	22.71
Fixed High	12.29	0.1305	0.4587	23.27
Fixed Mean	12.62	0.1303	0.4193	22.32
C3 / Middle-high	12.48	0.1302	0.4059	22.16
Late-high	12.92	0.1277	0.4259	21.65
Generic Linear	11.60	0.1276	0.4458	24.71

† Added exploratory controls. GT-256 is a matched reference bake, not a theoretical bound.

source resolution, atlas resolution, hole filling and eleven unseen evaluation cameras. Original geometry and UVs are retained; unseen cameras do not enter the bake. GT-256 uses the same source resolution, masks and bake parameters, with the same raw atlas coverage as each generated condition. Camera and sampling calibrations are documented in Supplementary Section S5. Evaluation uses unlit base color and does not measure PBR fidelity.

Table *[removed reference:* `tab:probe`*]* summarizes representative scheduling variants on the 24-object probe set. C4 and other late-stage high-scale strategies obtain higher ΔFG-SSIM, but they are accompanied by obvious decreases in Laplacian variance, indicating that their structural gains come at the cost of texture loss. This is the main reason a direct transfer of guidance scheduling intuition is insufficient: a schedule can make the condition stronger at apparently useful timesteps and still damage the texture signal that matters for 3D appearance. Uniform high-scale strategies show a similar problem. For example, $s = 2.50$ achieves ΔFG-SSIM of +0.078, but ΔLap Var decreases to −0.0108. In contrast, C3 achieves ΔFG-SSIM of +0.084. Although it is slightly lower than some aggressive variants in structure gain, its Laplacian variance loss is only −0.0006, close to zero texture loss. Therefore, C3 is the only variant labeled Promising. We freeze this configuration before large-scale evaluation and do not tune it on the 300-object setThe internally predeclared primary contrast was Late-high minus Fixed Mean on unseen-view FG-LPIPS, proposed from the earlier pilot and tested on the ten newly selected bake objects. Its mean is −0.00265 (95% object-bootstrap CI [−0.00414, −0.00137]; exact paired sign-flip $p = 0.00195$), with all ten objects favorable. This is an approximately 2.0% reduction relative to Fixed Mean's mean LPIPS and identifies a small allocation benefit within this sampling protocol. Generic Linear has a slightly lower mean LPIPS but lower PSNR; Late-high does not dominate every endpoint or control.

The specific configuration of C3 uses the conservative scale $s = 1.25$ in the early and late stages, and the strong scale $s = 2.50$ in the middle geometry-refinement stage, i.e. , C3 = (1.25, 2.50,1.25). Its motivation follows the staged role of diffusion denoising. In the early stage, latent variables are still highly noisy, and strong geometric correction can interfere with global structure formation. In the middle stage,

4.4.2. Reference gap and object-level effects

The matched GT-256 reference bake has mean LPIPS 0.0590, versus 0.1303 for Fixed Mean and 0.1277 for Late-high. Post hoc, the signal becomes more stable, making it suitable to strengthen geometric constraints such as normals, depth maps, and foreground contours. In the late stage, generation mainly focuses on color, material, and fine texture synthesis, so adapter intervention should be reduced to preserve the texture generation space of the base diffusion model.

Figure 2 provides a qualitative comparison between C3 Late-high change closes 3.7% of the mean Fixed Mean–GT reference gap (pointwise object-bootstrap CI [2.3%,5.3%]). This ratio of paired means is descriptive, not a theoretical ceiling or evidence of acceptable material fidelity. Leaving out each object in turn keeps the mean Late–Mean difference between −0.00294 and uniform adapter scaling. The red-boxed zoom-in regions make the intended comparison visible: conservative scaling preserves more color variation but provides weaker structural correction, whereas aggressive scaling strengthens geometric constraints but flattens surface texture. C3 applies strong geometric correction in the middle stage and reduces adapter intervention in the late stage, thereby preserving more local texture than $s = 2.50$ while retaining stronger geometric correction than the conservative scale. −0.00208, while the substantial reference-appearance discrepancies remain. Supplementary Section S5 provides the paired values and calculation.

Figure *[removed reference:* `fig:pareto`*]* further shows the distribution of the 17 variants in the shape-texture space. Most variants with high shape gain are accompanied by large texture loss and fall into the Trap region, whereas C3–Fixed High results from the separate earlier five-object pilot do not establish a stable C3 maintains a relatively high shape gain while keeping texture loss near zero. Thus, the value of benefit; complete endpoints and tests are in Supplementary Section S5. Late-high is a different allocation from C3is not that it maximizes every metric, but that it avoids the trap of "higher structural scores but flattened texture," resulting in a more stable shape-texture compromise.

Mechanistically, the advantage of C3 is not simply the result of hyperparameter tuning; it comes from the functional alignment between adapter correction and the diffusion process's stage-dependent objective. A natural hypothesis might be that correction should be strongest where the correction-to-noise ratio is highest. However, under the Euler sigma schedule, this ratio increases monotonically throughout denoising (from approximately 8 at full noise to over 400 near completion),which would predict that late-stage correction is most effective—a prediction inconsistent with the scheduling-variant results. The correct explanation involves *functional alignment* rather than signal-to-noise ratio. The adapter correction is derived from geometric conditions (normals, depth, contours) and is therefore constructive only when the diffusion process is actively refining geometric structure. In the early stage ($\sigma > 5$), global structure has not yet emerged, so the correction has no stable target to reinforce and instead disrupts layout formation. In the middle stage ($\sigma \approx 5$–1.2), coarse structure is established and the denoiser is performing geometric refinement—the adapter correction direction aligns with this objective, making it maximally constructive. In the late stage ($\sigma < 1.2$), generation shifts to color, material, and fine texture synthesis; geometric correction is orthogonal to this objective and competes with the texture formation process. C3's low-high-low schedule therefore allocates strong correction proportional to functional alignment, not to signal clarity. Notably, this allocation is the reverse of limited-interval CFG guidance [36]: under adapter residual injection, geometric correction must also be withdrawn near the end of sampling, because the late-stage risk is competition with texture synthesis rather than diversity loss. The schedule is tied to the denoising process's functional role at each stage rather than to validation-set-specific tuning or a generic monotonic guidance ruleGenerated-view and baked-view ranking associations and per-object winner changes are descriptive and remain in Supplementary Section S5; they do not establish a general ranking reversal. Figure 3 shows Tire, selected by an input texture property, and the explicitly post-outcome carved-font example, whose LPIPS improves while PSNR declines. The selected displays illustrate appearance

mismatch and an endpoint tradeoff, not prevalence or a general schedule benefit.

This explanation is consistent with the scheduling-variant results. Early high-scale or early-and-late high-scale variants provide limited structural gain, indicating that early strong intervention is unstable. The late-stage high-scale configuration C4 obtains a high structural score but shows a substantial decrease in Laplacian variance, making it a typical Trap variant where structural metrics improve while texture is flattened. Uniform high-scale variants similarly improve condition adherence but lose texture detail, showing that the issue is not solved by a stronger ControlNet-like scale. An aggressive C3 variant further increases the middle-stage strength but reduces signal fidelity, suggesting that even middle-stage correction has diminishing returns and can lead to over-constraint.

4.4.3. *Protocol and deployment boundaries*

Separate twenty-object GSO evaluation. The GSO/MV-Adapter study completes twelve conditions by three seeds on twenty fixed inputs, plus twenty matched GT self-bakes: 740 unique baked meshes and 8,140 unseen-view records. The objects, source meshes, projection cameras, texture resolution and eleven unseen cameras are shared across controls; unseen cameras do not contribute to baking.

Table *[removed reference:* `tab:extended`*]* compares representative variants under extended texture metrics. C3 has nearly zero loss in Laplacian variance and remains relatively stable in RGB standard deviation, gradient magnitude, and high-frequency energy. Compared with C4 and uniform high-scale strategies, C3 shows weaker texture degradation, indicating that its shape-texture balance is not an artifact of a single metric The original twenty-object bake protocol failed a discontinuous UV-ramp diagnostic after fifteen objects. A subsequent ground-truth-only calibration addressed finite-resolution mixing across UV boundaries and retained the fixed twenty-object cohort, cameras, masks, materials and resolutions. Because this correction followed inspection of initial outcomes, the resulting GSO evaluation is reported as protocol sensitivity. The original failure and remaining UV/coverage limits are detailed in Supplementary Section S4.

Late-high minus Fixed Mean unseen-view LPIPS is $+0.000084$ (pointwise interval $[-0.000103, +0.000286]$; original comparison-family Holm $p = 0.788$). This study does not detect the ten-object pilot's benefit. It changes both input domain and conditioning/projection protocol, so it is neither a direct same-protocol replication nor evidence assigning the difference to backbone alone. Table 5 reports the two separate bake primaries; their endpoint remains distinct from the FG-PSNR timing comparisons in Table 7.

Finally, as a transfer test rather than another search, we compare conservative uniform scaling ($s = 1.25$), aggressive uniform scaling ($s = 2.50$), and selected C3 (TCAS) on 300 objects. C3 is not searched again on the 300-object set; it directly uses the timestep schedule selected on the 24-object probe set to evaluate the transferability of the strategy to a larger object set. Because the 24-object probe is a subset of the 300-object pool, the main-table comparison is not probe-disjoint; a disjoint holdout re-verification is provided in Section *[removed reference:* `subsec:adapterdep`*]*. Supplementary Videos 1 and 2 accompany these bake studies. The GeoTex video shows two rights-cleared assets over eleven recorded unseen cameras; the GSO video shows one source-ordered object at seed 42 under twelve conditions and eleven protocol-specific cameras. The GSO source-mesh panel is not a GT self-bake. Object, camera, and display limits are detailed in Supplementary Sections S5 and S4.

Table 5. Unseen-view FG-LPIPS primary contrasts for the two bake studies.

Study (N)	Δ FG-LPIPS	95% CI	p
GeoTex (10)	−0.002652	[−0.004144, −0.001367]	0.00195
GSO (20)	+0.000084	[−0.000103, +0.000286]	0.788

Late-high minus Fixed Mean. GeoTex: exact paired sign-flip. GSO: original-family Holm16-adjusted paired sign-flip. Separate input and calibration protocols are not pooled; non-detection is not equivalence.

As shown in Table *[removed reference:* `tab:300obj`*]*, conservative uniform scaling $s = 1.25$ has lower FG-SSIM and Edge-SSIM, indicating insufficient geometric constraints. Aggressive uniform scaling $s = 2.50$ obtains the highest Edge-SSIM, suggesting that continuously strengthening adapter residuals helps edge and contour alignment. However, its PSNR is lower than that of C3, indicating that overly strong geometric intervention reduces overall signal fidelity. In contrast, C3 achieves an FG-SSIM of 0.476, which is comparable to $s = 2.50$, while increasing PSNR to 18.86 and slightly reducing FG-LPIPS. These results show that timestep-conditioned scaling preserves structural alignment while improving signal quality. The GSO20 self-bake and coverage provide empirical reference context; its complete ranking, coverage, and protocol-sensitivity analyses are in Supplementary Section S4. These quantities contextualize the method effects and do not define a theoretical quality bound.

PSNR mainly reflects global pixel error and should not be directly equated with texture detail quality . To further verify texture preservation, we compute Laplacian variance, RGB standard deviation, and gradient magnitude on the same 300 objects to measure foreground high-frequency response, color variation amplitude, and local gradient strength, respectively. All results use the same object list, random seed, 50 denoising steps, and adapter weights to ensure comparability

Native controls and protocol sensitivity. The added Native residual and No Adapter rows are post-inspection exploratory controls; their wrapper semantics and complete fifteen-test family are in Supplementary Section S5.

Table *[removed reference:* `tab:texture300`*]* shows that although $s = 2.50$ achieves the highest Edge-SSIM, it has the lowest values across all three texture metrics, indicating that full high-scale scaling weakens color variation and high-frequency details. A separate, previously studied fifty-object checkpoint comparison gives Late-high–Fixed Mean changes of -1.184 dB nonreference PSNR and $+0.00266$ nonreference LPIPS; C3 has FG-SSIM comparable to $s = 2.50$ while substantially improving Laplacian variance, RGB standard deviation, and gradient magnitude, demonstrating a better balance between structural alignment and texture preservationno corrected advantage over Fixed Mean or Fixed High. The changed checkpoint and protocol make this a sensitivity boundary, not an independent backbone confirmation (Supplementary Section S8).

As shown in Table *[removed reference:* `tab:retention`*]*, compared with conservative scaling $s = 1.25$, aggressive scaling $s = 2.50$ retains only 67.8% of Laplacian variance, 64.3% of RGB standard deviation, and 80.7% of gradient magnitude, indicating that full high-scale scaling substantially weakens foreground color variation and high-frequency details. In contrast, C3 retains 83.5%, 97.0%, and 88.3% of these metrics, respectively, and clearly outperforms $s = 2.50$ across all three texture measures. This shows that TCAS still incurs some texture reduction, but it substantially alleviates texture flattening compared with full high-scale scaling.



Fig. 3. GeoTex Bake10 appearance at held-out camera 004. Each row shows a GT-256 reference self-bake, Fixed Mean, and Late-high under a shared mesh/UV and within-object crop. Tire is selected by median source-foreground Laplacian variance among seven display-cleared assets. The carved font is an explicitly post-outcome example with the largest existing eleven-view Late-high minus Fixed Mean FG-LPIPS reduction among those seven (−0.00781); its FG-PSNR also decreases by 0.271 dB. The selected examples illustrate appearance mismatch and an endpoint tradeoff, not cohort prevalence or a general schedule benefit. Source images are 512×512 and displayed without method-specific color edits. The full ten-object effect distribution and scores are in Supplementary Section S5.

Native-pipeline quality boundary. On the same fifty inputs, the new geometry checkpoint trails full v29 by 2.16 dB FG-PSNR and has FG-LPIPS 0.0341 higher. This matched-input deployment comparison changes base weights and geometry implementation and uses a nondefault 50-step protocol, so it does not isolate TCAS. A same-base follow-up remains 0.0159 LPIPS worse than full v29 despite a local unscaled-versus-disabled geometry benefit. Neither result establishes TCAS deployment superiority; full tests and protocol details are in Supplementary Section S8.

Object-level statistics further show that C3 outperforms $s = 2.50$ on 87.7%, 95.0%, and 83.0% of objects in Laplacian variance, RGB standard deviation, and gradient magnitude, respectively.

4.5. Sensitivity Across Native Conditioning Interfaces

The auxiliary studies ask related scheduling questions under different native conditioning operators and camera contracts. The corresponding bootstrap 95% confidence intervals do not cross zero, indicating statistically stable texture-preservation advantages. C3 also outperforms $s = 2.50$ on 74.3% of objects in PSNR. By contrast, the confidence interval for the FG-SSIM difference includes zero, so these data do not establish a significant difference in foreground structural similarity between C3 and $s = 2.50$. Therefore, TCAS does not noticeably sacrifice structural alignment, while significantly reducing the texture degradation caused by full high-scale scaling.

In summary, the 300-object experiments demonstrate that adapter scaling affects both structural alignment and texture detail preservation. Full high-scale scaling strengthens edge constraints but is more likely to cause texture flattening. TCAS strengthens geometric correction in the middle stage and reduces adapter intervention in the late stage, thereby maintaining structural alignment while substantially reducing texture loss. It should be noted that C3 still has some texture decrease relative to $s = 1.25$; therefore, we do not claim that TCAS completely eliminates texture loss, but rather that it significantly reduces texture degradation compared with full high-scale scaling.

To address the concern that proxy texture statistics do not directly measure perceptual quality, we additionally evaluate CLIP-IQA *[removed citation:* `wang2023clipiqa`*]*, which builds on CLIP image–text representations *[removed citation:* `radford2021clip`*]*, on a 50-object subset of the evaluation pool as a complementary no-reference perceptual signal. CLIP-IQA is used as complementary perceptual evidence rather than a complete replacement for human preference evaluation.

Table *[removed reference:* `tab:clipiqa`*]* shows that C3 achieves a foreground CLIP-IQA score of 0.4906 versus 0.4860 for $s = 2.50$, scoring higher on 47 of the 50 objects (94.0%). The paired comparison uses a two-sided paired t-test over objects ($t(49) = 10.52$, $p \approx 3.6 \times 10^{-14}$), and the 95% confidence interval of the paired mean difference is constructed by object-level percentile bootstrap with 10,000 resamples, giving [0.0037, 0.0054], which excludes zero. The ordering $s = 1.25 >$ C3 $> s = 2.50$ supports the diagnosis that aggressive scaling reduces perceptual quality, while TCAS recovers most of this loss with stronger geometry than conservative scaling.

To further examine whether the diagnostic texture differences are perceptually meaningful, we conducted a blinded three-alternative forced-choice (3AFC)preference study on 30 objects drawn from the evaluation pool. Twenty-four participants viewed, for each ob-

ject and each of the three criteria (texture naturalness, shape consistency, overall quality), the anonymized outputs of conservative uniform scaling, aggressive uniform scaling, and C3 side by side; the left-to-right order of the three conditions was randomized independently per trial, GSO100 studies both use MV-Adapter; GSO100 is a separate source-ordered cohort and protocol sensitivity, not a third independent backbone. MVDiffusion scales a partial correspondence operator and is reported only as additional context. Table 7 presents each study's own specified primary contrast: the GeoTex contrast compares Middle-high with the mean of Early-high and participants were not informed which method produced which image or how many conditions existed. Instructions asked participants to select the best image for the stated criterion only. Each participant judged all 30 objects for all three criteria, yielding 720 valid first-choice decisions for each criterion. The study complements the proxy diagnostics by testing whether the observed texture and structure differences are reflected in human preference. Late-high, whereas both MV-Adapter contrasts compare Middle-high with Fixed Mean. We report these rows separately and do not pool them or treat them as an identical replication.

The preference pattern in Table *[removed reference:* `tab:preference`*]* follows the intended trade-off. Conservative scaling is most often selected for texture naturalness, while aggressive scaling is most often selected for shape consistency. C3 stays close to both single-scale baselines on their respective strengths and receives the highest overall preference, with 58.1% of all overall-quality votes. Thus, TCAS is not claimed to maximize every individual criterion; it provides a perceptually preferred balance between geometric adherence and texture preservation

Input-selected native MV-Adapter study. We selected fifty Fresh C inputs not previously evaluated through MV-Adapter, using input availability and native material/geometry compatibility. Native mesh normalization and six orthographic cameras define both generation and evaluation targets. Three-object identity checks confirmed agreement between the deployed and recorded default paths. An earlier seven-object panel had an invalid camera calibration and is excluded from these comparisons (Supplementary Section S3).

Because first-choice decisions are clustered within both participants and objects, we additionally report 95% confidence intervals obtained by a two-level cluster bootstrap (resampling participants and objects with replacement, 10, 000 resamples). C3's overall-quality share is 58.1% The seven conditions share object inputs, seed, geometry and evaluation code. With 50 steps, low/high are 0.75/1.0; stage-high schedules contain 16 high steps, Fixed Mean is 0.83, and Generic Linear is 0.75–0.91. These five controls have matched per-feature sums within this interface. All 50 objects remain in the primary cohort, including thirteen texture-absent sources with legal constant-white base-color targets. Targets are unlit native-camera base color, not Fresh C's differently normalized reference renders or certified physical albedo.

The internally predeclared primary, Middle-high minus Fixed Mean FG-PSNR, is +0.029 dB (95% CI [50.1%, 65.8%]), exceeding the conservative and aggressive baselines by +33.6 (CI [19.3, 47.2])and +40.6 percentage points (CI [29.4, 51.8]); both differences exclude zero, so the overall-preference advantage of object-bootstrap CI $[-0.019, +0.081]$; paired Monte Carlo sign-flip $p = 0.281$; 29/50 favorable). This does not support a stable middle-window gain or establish equivalence. The remaining 104 pair–metric tests use one Holm family. Middle-high has lower Edge-SSIM than Fixed Mean (adjusted $p = 0.00104$); some other placement contrasts

Table 6. Native-camera MV-Adapter means on fifty input-selected objects.

Condition	FG-PSNR	FG-LPIPS	FG-SSIM	Edge-SSIM
Native / Fixed High	12.60	0.1280	0.5041	0.3526
Fixed Low	12.73	0.1270	0.5066	0.3475
Fixed Mean	12.68	0.1279	0.5068	0.3509
Early-high	12.60	0.1296	0.5041	0.3503
Middle-high	12.71	0.1262	0.5061	0.3470
Late-high	12.74	0.1260	0.5075	0.3514
Generic Linear	12.72	0.1263	0.5072	0.3501

separate, including Late-high versus Early-high PSNR. Such endpoint-dependent differences do not demonstrate transferable C3 is significant under cluster-aware inference. By contrast, optimality. Complete contrasts, correction rules and the textured subset are in Supplementary Section S3. This is a native-interface validation of the baselines' single-criterion advantages are not significant (texture naturalness: $s = 1.25$ over C3 +6.7 points, CI $[-7.9, 21.5]$; shape consistency: $s = 2.50$ over C3 +5.3 points, CI $[-6.7, 17.9]$). Overall quality is therefore the only criterion with a significant preference, supporting the interpretation of TCAS as the perceptually preferred balance rather than a per-criterion maximum.

scheduling question, not architecture-matched replication of GeoTex's capped residual intervention.

TCAS uses a fixed three-stage scale, which has the advantages of requiring no training, being simple to implement, and introducing low computational overhead. However, its residual scaling is still uniform across spatial positions and frequency components, lacking fine-grained adaptivity. To examine whether stage-wise adapter control can be extended to learned adaptive correction, we trained the FAC extension—LTAG for timestep gating, GSG for spatial gating, and FSC for frequency-selective correction on top of the TCAS temporal schedule. The base model, VAE, and GeoTex-Adapter remain frozen ; only the three lightweight controllers are trainable, adding roughly 6×10^3 parameters. The controllers are trained with the same enhanced denoising objective as the base adapter—a latent noise-prediction MSE with 1.7× foreground and 1.3× edge spatial reweighting plus a latent-SSIM term—using AdamW (peak learning rate 10^{-4}, 100 warm-up steps) for 2,000 steps on the adapter's training pool, which is strictly disjoint from all evaluation objects, and all variants are evaluated on the strong-residual, per-layer-capped adapter instance of Section *[removed reference:* `subsec:adapterdep`*]* under a strictly paired protocol (identical objects, seeds, and evaluation code)

Separate external-input MV-Adapter sensitivity. The source-ordered GSO100 convenience cohort uses the same native MV-Adapter control path and three shared seeds. It includes seven scale/time profiles, with four inherited profiles frozen after earlier baseline exposure. The scheduling comparison is an external protocol sensitivity, not an independent prospective confirmation or parameter search.

In this controlled setting, no learned variant reached the training-free schedule. We tested seven configurations spanning the most favorable training conditions we could construct: joint training of all controllers; a two-stage scheme with the temporal gate warm-started at the exact C3 envelope and frozen while the spatial and frequency controllers train; an envelope penalty that constrains the gate to remain near C3 during joint training; a fidelity-weighted objective; and a gently scheduled variant with fourfold-reduced training. Their paired deficits relative to the TCAS baseline range from −1.1 to −2.9 dB foreground PSNR (FG-PSNR; win rates at most 16%), increasing monotonically with the amount of controller training. A warm-start control in which the gate is left untrained at the exact C3 envelope is statistically indistinguishable from TCAS (The specified all-six-view Middle-high minus Fixed Mean primary is −0.014 dB (interval $[-0.031, +0.004]$; paired $p = 0.122$). A secondary Late-

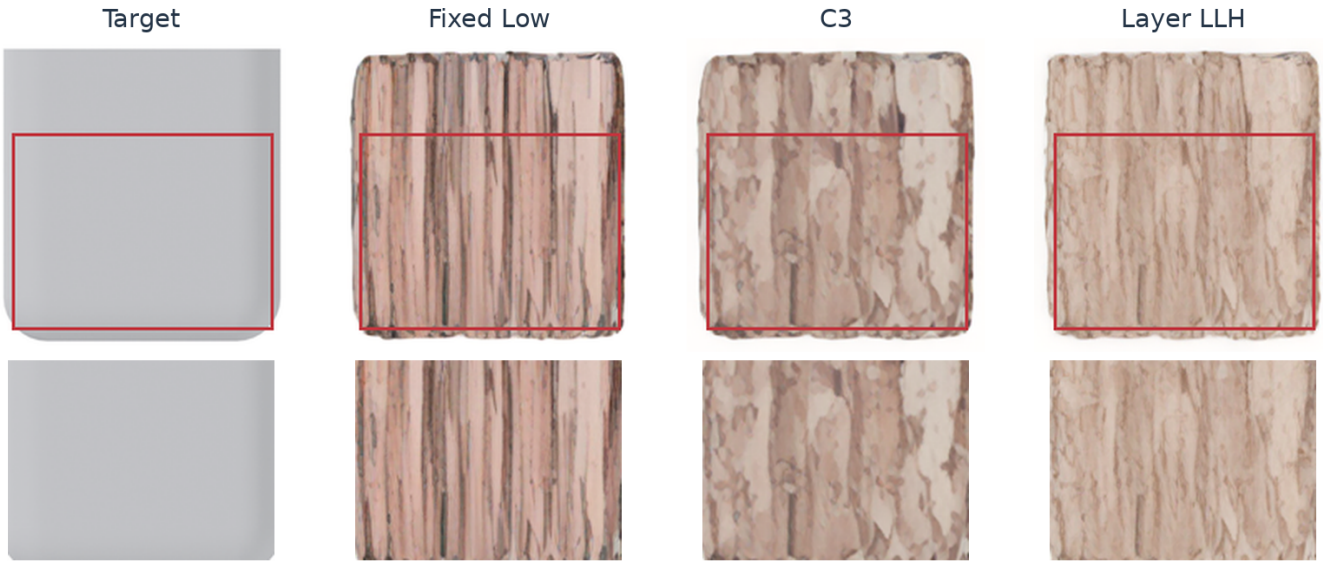


Fig. 4. Fresh C view 0, E5-5-2: Target, Fixed Low, C3 and layer LLH share a complete-object frame and an identical pixel crop. The generated brown pattern does not reproduce the gray target; changing its variation does not establish color fidelity. This previously viewed example came from an existing C3–GFL LPIPS stratum and was not selected for color. It does not reproduce the legacy purple-shift cases. Object credits are supplied in Supplementary Section S9.

high–Fixed Mean contrast is +0.045 dB ([+0.024, +0.065]; Holm83 $p = 0.00335$), with SSIM +0.00134 (adjusted $p = 0.000415$). Its five-nonreference-view LPIPS difference is −0.000086 ([−0.000551, +0.000413]; separate Holm21 $p = 1$). Thus a small detected image effect does not establish a perceptual or baked-appearance benefit. Table 7 reports the original Fresh C timing primary first, followed by the two study-specific native-interface FG-PSNR $\Delta = -0.18$ dB on the 12-object control, 95% CI [−0.42, +0.05], crossing zero), validating the evaluation path. No configuration or hyperparameter was selected on any evaluation object; the complete dose–response study over all seven configurations is provided in the supplementary materialprimaries; Table 5 retains the separate bake LPIPS primaries. Full paired secondary families remain in Supplementary Sections S3–S5. The complete rights-cleared native-camera comparison across all seven MV-Adapter conditions is retained in Supplementary Section S3.

For transparency, we note that an initial evaluation of the FAC extension, conducted on the main adapter under our earlier protocol before the strictly paired and strictly disjoint training controls were adopted, had suggested a positive gain. That initial comparison differed in both evaluation protocol and adapter instance, and it does not meet the controls we now consider necessary for a transfer claim; under the strictly paired, strictly disjoint protocol above, no trained configuration replicates the gain. We therefore report only the controlled re-examination and treat the initial comparison as superseded

Fixed-seed sensitivity. A post-inspection analysis retains all three recorded sampling seeds. GSO100 Late–Mean PSNR averages are positive at each seed (+0.057, +0.038, +0.039 dB), but only 27/100 objects improve at all three and 68/100 have mixed or tied directions. Its nonreference LPIPS averages change direction (−0.000389, −0.000269, +0.000398). For the GSO20 bake, only 1/20 objects has lower Late–Mean LPIPS at all three seeds; 14/20 have mixed or tied directions. Sign agreement has no practical-effect threshold. These conditional, object-level summaries do not estimate training randomness or revise the original primary tests; full seed intervals and deltas are in Supplementary Section S4.

The outcome delimits the scenario in which learned modulation affects the paper's conclusions. Preserving the temporal envelope is necessary but not sufficient: with the envelope preserved and the controllers untrained, the extension neither helps nor harms, while every trained configuration degraded fidelity, indicating that the remaining damage is driven by the learned-modulation training signal itself, which trades foreground fidelity for denoising-objective loss. Learning a modulation that improves upon the C3 schedule therefore requires a training signal that does not trade away foreground fidelity (e.g., fidelity-weighted objectives, envelope-preserving constraints, or post-hoc rather than gradient-learned gates), which we identify as future work. With the tested lightweight controllers, the fixed schedule is the stronger default, and TCAS, which is training-free and plug-and-play, remains the main and sole method of this paperretrospective 75-object MVDiffusion panel scales correspondence-block outputs rather than the GeoTex residual operator. Its layer-wise L–LLH condition is worse than global fixed low in PSNR, structural agreement and reference-color error; LPIPS does not separate. This result and the earlier MV-Adapter panel remain in Supplementary Section S3 and delimit the scope of cross-interface transfer.

Projection-guidance candidates are retained as exploratory negative results in Supplementary Sections S4 and S8; they do not form a validated alternative method.

4.6. *Historical Cohort Sensitivity and Prior Evaluation*

The preceding experiments demonstrate the texture-flattening failure mode under high adapter scaling on the studied adapter checkpoint. A natural question is whether this failure mode is universal or specific to the adapter's trained residual properties. We conduct additional analysis across adapter checkpoints with different residual magnitudes to characterize how the failure mode depends on adapter properties. historical +0.96 dB estimate has not been reproduced under an authenticated, matched protocol. The exposed 276-request retraining is a protocol/checkpoint sensitivity panel, not a verified 276-object holdout: one failed input was randomly replaced and

Table 7. Study-specific FG-PSNR schedule primaries.

Study	Contrast	N	Δ FG-PSNR (dB)	95% CI (dB)	p
GeoTex Fresh C	Middle − mean(Early, Late)	300	−0.033877	[−0.117170, +0.051422]	0.432378
MV-Adapter50	Middle − Fixed Mean	50	+0.028986	[−0.019411, +0.080739]	0.280747
GSO100	Middle − Fixed Mean	100	−0.013660	[−0.030565, +0.003519]	0.122279

Middle/Early/Late denote the corresponding high-window allocations. GeoTex tests Middle against mean(Early,Late); each MV-Adapter study tests Middle against Fixed Mean. GSO100 uses the same MV-Adapter backbone as the 50-object study. Pointwise paired-object intervals; equal applied sums do not imply equal realized residual dose.

its actual UID was not recorded. C3–Fixed High reverses by endpoint (+1.699 dB full-frame PSNR; −1.314 dB FG-PSNR), and the panel lacks Fixed Mean and moved-window controls, so it does not identify a timing effect. The original 3AFC applies only to its thirty historical stimuli, and FAC results retain incomplete run and checkpoint identity. Their detailed values, provenance and limits remain in Supplementary Section S7; none validates the current cohort's preference or baked fidelity.

We find that the specific failure mode of aggressive adapter scaling is not universal. Across three adapter checkpoints spanning a spectrum of trained residual magnitudes, uniform high scale ($s = 2.50$) produced: (a) texture flattening when all layers are scaled without per-layer caps; (b)high-frequency artifact amplification when per-layer caps restrict the fine-texture layer to a low scale while coarse layers are boosted; or (c) no appreciable effectwhen the adapter's residual magnitude is very small. A per-layer scale-cap ablation on the same checkpoint reproduced both (a) and (b), isolating the inference-time scale semantics as the determining factor for the failure *direction*.

5. Discussion and Limitations

A residual-magnitude sweep further shows that the intervention *strength* is monotonic in

Why nominal control schedules can mislead. The principal contribution is an evidence chain from the trained residual magnitude: scaling the adapter's output projection from 0.1× to 3× transitions the behavior from no-effect, to artifact amplification under high scale, to over-constraint even under conservative scale.

Moreover, a six-object stage ablation shows that even *which denoising stage is most sensitive to high scale* is adapter-dependent: on the artifact-prone adapter, early-stage high scale reproduced nearly all of the damage (FG-SSIM ↓ from 0.266 executed residual interface to generated views and the final textured asset, rather than a new schedule parameterization. This distinction matters because C3's gain over Fixed Low changes both strength and timing: the deep and middle applied-scale sums rise from 62.5 to 0.187) while late-stage high scale was nearly harmless (FG-SSIM 0.261), whereas on the paper's main adapter the late stage was the primary source of texture loss.

Despite this variability, in every regime where scaling had a non-negligible effect, the temporal low-high-low schedule 82.5, while the shallow requests are clipped to the same 0.8 value. Fixed High and Generic Linear have better means than C3 yielded the best shape-texture balance. This indicates that TCAS's value is structural rather than incidental: by relying on the diffusion process's stage-dependent roles (global layout → geometry refinement → texture synthesis), it mitigates the harm of high-scale intervention across the tested adapter regimes, despite their adapter-specific failure modes. The failure mode (flatten vs. artifact vs. no-opacross all four image endpoints. With per-group applied sums matched, the specified Middle-high contrast is −0.0339 dB (95% interval [−0.1172, +0.0514]), its strength, providing no detected middle-window advantage. A separate retrospective all-pair family distinguishes Middle-high from Early-high and Late-high in both FG-PSNR and its temporal location are all adapter-dependent; what remains stable in our experiments is that the middle denoising stage was the best-performing tested window for geometric correction, although whether this holds across adapter families remains an open question for future work. FG-LPIPS, indicating placement sensitivity within this cohort. Equal scalar sums still allow the adapter activations and hidden-state trajectory to change, so these contrasts identify an allocation effect at matched nominal budget, not a residual-dose mechanism or a universally preferred late schedule. Reporting the actual cap-adjusted intervention and testing strong alternatives therefore changes how the original C3 result should be interpreted.

Two further robustness checks were performed on the strong-residual, per-layer-capped adapter instance described above. First, a boundary/peak sensitivity sweep (three high-scale windows [0.25, 0.75), [1/3, 2/3), [0.4, 0.6) × five peak scales from 2.00 to 3.00, fifteen candidates on the 24-object probe protocol) shows that the canonical middle-third window lies in the stable high-performance region: its 2.50 setting achieves 16.71 dB whole-image PSNR, the best setting in the same window (2.75) achieves 16.79 dB with worse texture diagnostics and lower FG-SSIM, and the broader and narrower windows reach at most 16.73

What survives texture baking. Generated-view and 16.62 dB. C3 is therefore a robust middle-stage choice rather than a boundary-sensitive point estimate. Second, because the 24-object probe is part of the 300-object pool, transfer claims were re-verified on the disjoint 276-object holdout (obj_{0024}–obj_{0299}), summarized in Table *[removed reference:* `tab:holdoutsched`*]*: C3 improves PSNR over the strongest non-C3 probe candidates by +0.258 dB (95% CI [0.207, 0.308], 206/276 wins) over a trapezoid schedule and +0.324 dB ([0.281, 0.367], 232/276 wins) over a Gaussian-peak schedule , and over the reviewer-relevant generic schedules by +0.323 dB ([0.263, 0.386], 214/276 wins, FG-SSIM statistically tied) over linear warm-up and +0.902 dB ([0.824, 0.981], 253/276 wins, FG-SSIM +0.046) over the cosine bump. The holdout ordering matches the probe: warm-up approaches C3 in structure but loses in signal fidelity, and baked-view rankings are positively associated in the GeoTex pilot, but projection and texture construction remain consequential: point-estimate winners can change between endpoints. On ten compatibility-selected objects, Late-high reduces unseen-view LPIPS by 0.00265 (about 2% relative to Fixed Mean) and closes 3.7% of the reference self-bake gap. This is a small local asset-level effect beside substantial remaining appearance error, not a texture-quality upper bound or evidence of faithful material reproduction. The separate GSO20 bake study does not detect the same benefit after a calibration change made following initial outcome inspection; its different cohort and protocol prevent treating it as a direct

replication. Likewise, the smooth bump loses in both because its effective high-scale duration is shorter than the full middle third.

new geometry checkpoint trails full v29 by 2.16 dB FG-PSNR and has FG-LPIPS higher by 0.0341 under a recorded non-default matched-input comparison. Because weights and geometry implementations differ, this marks a deployment-quality boundary rather than a schedule effect. Unlit base-color renders and incoming-view variance do not establish PBR fidelity, seams, or physical albedo.

This paper mainly focuses on inference-time adapter scaling and does not retrain the base multi-view diffusion model or the adapter. Therefore, TCAS can alleviate texture degradation under high scales—which may manifest as flattening or artifact amplification depending on the adapter's residual properties—but it cannot fundamentally solve the limitations of the base model in invisible-region completion, complex material synthesis,

Interface and object dependence. The native MV-Adapter50 and cross-view detail consistency.

Second, Laplacian variance, RGB standard deviation, and gradient magnitude are proxy metrics for texture degradation. They are effective for detecting high-frequency loss and surface smoothing, but they cannot fully replace human perceptual evaluation. For objects with strong reflection, transparency, thin structures, or highly stylized textures, discrepancies may exist between these metrics and subjective quality. GSO100 primaries do not detect their specified middle-window contrasts, but their comparators differ from the GeoTex timing primary, and GSO100 uses the same MV-Adapter interface rather than an independent backbone. MVDiffusion tests only a partial correspondence operator. These studies delimit transfer across implementations; they do not form one pooled or architecture-matched replication. A retrospective FRESH_CONFIRM_300 analysis also associates foreground texture variation with the LLH–Fixed Low response (Laplacian-variance Spearman $\rho = -0.524$; quartile mean effects $+4.140, +1.127, -0.087, -0.029$ dB). That association identifies a candidate source of response heterogeneity, not a causal texture effect, threshold, or adaptive schedule. Together, these results give graphics researchers a reason to report the interface, camera contract, and object-level asset response when evaluating geometry-control allocation.

In addition, the current TCAS uses fixed stage boundaries and a fixed scale combination, C3 = (1.25, 2.50, 1.25). This setting is simple and stable, but it should be understood as a validated inference strategy for the studied geometry-adapter pipeline rather than a universally optimal schedule. As shown in Section *[removed reference:* `subsec:adapterdep`*]*, the failure mode of high-scale intervention is adapter-dependent, and even the most sensitive denoising stage differs across adapters; yet

Practical implications and limitations. For a texturing system, the evidence supports comparing the deployed control with strong within-interface settings, reporting requested and applied profiles, and evaluating shared-mesh bakes from unseen cameras against a matched reference self-bake. It does not justify replacing a deployed fixed setting with C3 's middle-stage concentration was the best-performing schedule among all tested variants in every regime examined, suggesting that its temporal structure is robust even when specific failure mechanisms vary. It has not yet been adapted to object complexity, material type, input reference image quality, or different sampler families, and the schedule has not yet been validated on a different geometry-conditioned multi-view diffusion backbone. Future work could study content-adaptive stage boundaries, SNR-aware scaling, and learned scale selection strategies.

Finally, our controlled re-examination of the learned FAC extension indicates that unconstrained lightweight gating can flatten the temporal envelope that makes TCAS effective; learned modulation is therefore promising only under envelope-preserving constraints , which we identify as a concrete direction for future work, while TCAS remains the main training-free methodor selecting Late-high universally. The bake cohorts are compatibility-selected, the GSO calibration followed initial outcomes, and the matched-budget analysis is retrospective on an already evaluated cohort. Fresh C uses controlled retraining rather than an authenticated historical main checkpoint; the 276-slot panel includes an unknown replacement UID and does not verify distinct held-out coverage. Human preference applies only to its original stimuli, historical FAC checkpoints are incomplete, and three fixed seeds do not estimate training variation. These constraints limit replication and transfer while leaving the paired within-protocol results interpretable; detailed records remain in Supplementary Sections S6–S8.

6. Conclusion

This paper presents a systematic experimental analysis of adapter scaling in multi-view diffusion texture generation . The experiments show that higher adapter scales continuously improve structural metrics such as FG-SSIM, but such improvement can mask texture quality degradation. In the 26-object texture audit, 92% of objects show texture flattening and high-frequency detail loss under $s = 2.50$. This reveals an adapter-specific temporal conflict that is not captured by simply treating the scale as a generic guidance strength.

To address this shape-texture trade-off, we propose Timestep-Conditioned Adapter Scaling (TCAS). The selected configuration, C3, applies strong adapter correction in the middle denoising stage and uses conservative scales in the early and late stages. The 24-object probe set is used only to select C3 from 17 scaling variants that include uniform, layer-wise, and temporal alternatives. With this schedule fixed, the 300-object evaluation pool further shows that We evaluated geometry-control allocation from its implemented GeoTex residual interface through image generation and native-UV texturing. On 300 held-out objects, stronger fixed and linear controls revise the interpretation of C3achieves FG-SSIM comparable to aggressive uniform scaling $s = 2.50$ while improving PSNR by 0.96 dB's gain over Fixed Low, while the specified matched-budget primary detects no middle-window advantage. Baking shows a small Late-high unseen-view LPIPS improvement on ten compatible objects alongside substantially larger reference-appearance error; a separate bake protocol does not detect the same benefit. These findings make the graphics contribution an implementation-aware, asset-level evaluation of control allocation, rather than a new universal scheduling method. They support comparing applied controls with strong alternatives and measuring the textured asset from unseen views, without identifying a universally preferred schedule or establishing faithful material reproduction.

Therefore, the core contribution of this paper is not to claim that the generated results are visually close to the ground truth in all cases, nor to present TCAS as a universal replacement for all guidance schedules. Instead, the contribution is to reveal and validate a hidden shape-texture trade-off caused by geometry-adapter residual injection, and to show that a simple non-monotonic inference schedule can mitigate this trade-off better than fixed high-scale control and other tested scheduling variants. This conclusion suggests that, in the geometry-controlled multi-view pipeline studied here—and plausibly in similar geometry-conditioned texture pipelines—adapter strength is better treated as a stage-dependent inference control variable than as a single fixed hyperparameter.

As an additional check, a controlled re-examination of the learned-correction extension (FAC) under a strictly paired protocol showed that unconstrained lightweight gating does not preserve the schedule's shape; the analysis delimits the scenario in which learned modulation helps—only under envelope-preserving constraints—and the training-free TCAS remains the main and sole method.

Data availability

All data included in this study are available upon request by contact with The accompanying research package contains object-level metrics, frozen protocols and configuration records, analysis scripts, and the LaTeX source needed to rebuild the manuscript and Supplement. Recovered FAC source is supplied with a record of its entry points and validation scope; these component checks do not constitute a historical full-model replay. Statistical summaries can be recomputed without model weights, while fresh inference requires the recorded weights, runtime and licensed source assets. Raw generated-image archives, trained checkpoints and participant-level responses are not included. Source assets must be acquired under their recorded licenses. Additional analysis records may be requested from the corresponding author, subject to licensing, privacy and archive constraints.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] ControlNet project. Official depth-to-image demo: layer-dependent control scales. Source code, fixed commit f2e1bc38dc8d925fb542be4c138c46013ace1988, 2023. `https://github.com/lllyasviel/ControlNet/blob/f2e1bc38dc8d925fb542be4c138c46013ace1988/gradio_depth2image.py`. Accessed 12 October 2026.

[2] Hugging Face. Diffusers v0.21.0: ControlNet conditioning scale and guidance start/end. Official versioned documentation, 2023. `https://huggingface.co/docs/diffusers/v0.21.0/api/pipelines/controlnet`. Accessed 12 October 2026.

[3] Kulkarni A S. Scheduled Style Injection: Expanding the Style-Content Pareto Frontier in Training-Free Diffusion-Based Style Transfer. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), NTIRE Workshop*, 2026.

[4] Liu R, Wu R, Van Hoorick B, et al. Zero-1-to-3: Zero-shot one image to 3D object. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[5] Shi R, Chen H, Zhang Z, et al. Zero123++: A single image to consistent multi-view diffusion base model. arXiv:2310.15110, 2023.

[6] Liu Y, Lin C, Zeng Z, et al. SyncDreamer: Generating multiview-consistent images from a single-view image. *International Conference on Learning Representations*, 2024.

[7] Long X, Guo Y C, Lin C, et al. Wonder3D: Single image to 3D using cross-domain diffusion. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[8] Shi Y, Wang P, Ye J, et al. MVDream: Multi-view diffusion for 3D generation. *International Conference on Learning Representations*, 2024.

[9] Wang P, et al. ImageDream: Image-prompt multi-view diffusion for 3D generation. arXiv:2312.02201, 2023.

[10] Xu J, Cheng W, Gao Y, Wang X, Gao S, Shan Y. InstantMesh: Efficient 3D mesh generation from a single image with sparse-view large reconstruction models. arXiv preprint arXiv:2404.07191, 2024.

[11] Li P, Liu Y, Long X, et al. Era3D: High-resolution multiview diffusion using efficient row-wise attention. *Advances in Neural Information Processing Systems*, 2024.

[12] Perla S R K, Zhang H, Mahdavi-Amiri A. Advances in neural 3D mesh texturing: A survey. *Computer Graphics Forum*, 2026, e70392.

[13] Chen D Z, et al. Text2Tex: Text-driven texture synthesis via diffusion models. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[14] Richardson E, Metzer G, Alaluf Y, et al. TEXTure: Text-guided texturing of 3D shapes. *ACM SIGGRAPH 2023 Conference Proceedings*, 2023.

[15] Poole B, Jain A, Barron J T, Mildenhall B. DreamFusion: Text-to-3D using 2D diffusion. *International Conference on Learning Representations*, 2023.

[16] Chen R, Chen Y, Jiao N, Jia K. Fantasia3D: Disentangling geometry and appearance for high-quality text-to-3D content creation. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[17] Guzmán J E, Mould D, Paquette E. Adaptive multiresolution exemplar-based texture synthesis on animated fluids. *Computers & Graphics*, 2026, 134: 104490.

[18] Lv X, Wu Z, Xu J, et al. Interpretable procedural material graph generation via diffusion models from reference images. *The Visual Computer*, 2025, 41(13): 11195–11205.

[19] Zeng X, Chen X, Qi Z, et al. Paint3D: Paint Anything 3D with Lighting-Less Texture Diffusion Models. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

[20] Cheng W, Mu J, Zeng X, et al. MVPaint: Synchronized multi-view diffusion for painting anything 3D. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025.

[21] Zhao Z, Lai Z, Lin Q, et al. Hunyuan3D 2.0: Scaling diffusion models for high resolution textured 3D assets generation. arXiv:2501.12202, 2025.

[22] Shao M, Xiong F, Sun Z, Xu M. MVPainter: Accurate and detailed 3D texture generation via multi-view diffusion with geometric control. arXiv:2505.12635, 2025.

[23] Huang Z, et al. MV-Adapter: Multi-view consistent image generation made easy. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025.

[24] Tang S, Zhang F, Chen J, Wang P, Furukawa Y. MVDiffusion: Enabling holistic multi-view image generation with correspondence-aware diffusion. *Advances in Neural Information Processing Systems*, 2023.

[25] Liu Y X, Xie M S, Liu H Y, Wong T T. Text-guided texturing by synchronized multi-view diffusion. *SIGGRAPH Asia 2024 Conference Papers*, 2024.

[26] Xu Q, Han X G, Zhang L. WonderTex: Consistent-and-seamless texture generation with text-guided multi-view image diffusion models. *IEEE Transactions on Visualization and Computer Graphics*, 2026, 32(7): 5815–5825.

[27] Mou C, Wang X, Xie L, et al. T2I-Adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024.

[28] Zhang L, Rao A, Agrawala M. Adding conditional control to text-to-image diffusion models. *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

[29] Ye H, Zhang J, Liu S, Han X, Yang W. IP-Adapter: Text compatible image prompt adapter for text-to-image diffusion models. arXiv:2308.06721, 2023.

[30] Hu E J, Shen Y, Wallis P, et al. LoRA: Low-rank adaptation of large language models. *International Conference on Learning Representations*, 2022.

[31] Liu S Y, Wang C Y, Yin H, et al. DoRA: Weight-decomposed low-rank adaptation. *International Conference on Machine Learning*, 2024.

[32] Zhang Q, Zuo S, Liang C, et al. AdaLoRA: Adaptive budget allocation for parameter-efficient fine-tuning. *International Conference on Learning Representations*, 2023.

[33] Ho J, Jain A, Abbeel P. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 2020.

[34] Song J, Meng C, Ermon S. Denoising diffusion implicit models. *International Conference on Learning Representations*, 2021.

[35] Ho J, Salimans T. Classifier-free diffusion guidance. *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, 2021; arXiv:2207.12598, 2022.

[36] Kynkäänniemi T, Aittala M, Karras T, et al. Applying guidance in a limited interval improves sample and distribution quality in diffusion models. *Advances in Neural Information Processing Systems*, 2024.

[37] Castillo A, Kohler J, Pérez J C, et al. Adaptive guidance: Training-free acceleration of conditional diffusion models. *Proceedings of the AAAI*

Conference on Artificial Intelligence, 2025.
[38] Chung J, Hyun S, Heo J P. Style Injection in Diffusion: A Training-free Approach for Adapting Large-scale Diffusion Models for Style Transfer. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
[39] Wang Z, Bovik A C, Sheikh H R, Simoncelli E P. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 2004, 13(4): 600–612.
[40] Zhang R, Isola P, Efros A A, Shechtman E, Wang O. The unreasonable effectiveness of deep features as a perceptual metric. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, 586–595.